



AS7056/57 Chip Library

API Documentation
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1 Introduction

This document provides an overview on how the vital signs sensor family AS7056/57 can be controlled through the ChipLib (Chip Library) and the corresponding OSAL (operating system abstraction layer).

The ChipLib's API allows for usage in several different fields of application. It is programmed in standard C language.

Key features:

- Driver library for convenient configuration and measurement
- Independent of hardware and platform
- Source code tested by ams

Limitations:

- Sensor calibration is not covered by this library.
- The library provides only raw values.

1.1 Overview

The ChipLib can directly be called by the user application and can be used without adaption. Hardware- and platform-dependent aspects must be handled in the OSAL. The OSAL interface is kept simple to keep integration efforts for the customers to a minimum. More information on the API functions can be found in the section [OSAL Functions](#). Details on how to implement application-specific OSAL are provided in the section [OSAL](#).

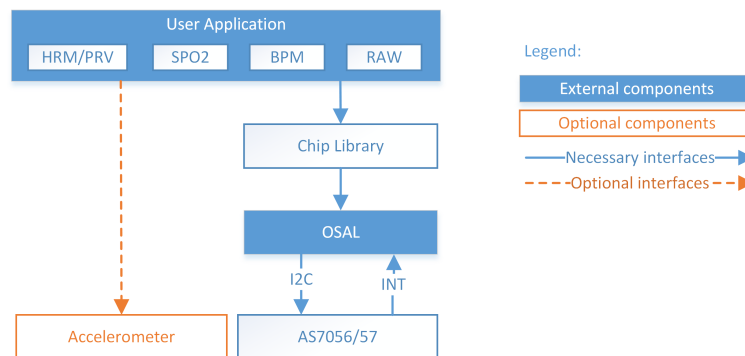


Figure 1 High-Level Block Diagram

- Accelerometer: Separate component providing accelerometer measurements, which are required by the HRM application for best performance.
- AS7056/57: Physical sensor device.
- OSAL: Hardware- and platform-dependent aspects of ChipLib.
- Chip Library: Control logic for the sensor.
- User Application: Signal processing applications to calculate vital signs data like HRM, SpO2 etc.

1.2 Acronyms and Abbreviations

1.2.1 Acronyms

API = Application Programming Interface
 BPM = Blood Pressure Monitor
 HRM = Heart Rate Monitor
 I2C = Inter-Integrated Circuit
 OSAL = Operating System Abstraction Layer
 SpO2 = Oxygen Saturation

1.2.2 Abbreviations

ChipLib = Chip Library

2 ChipLib

This page describes how to work with the ChipLib in more detail.

2.1 Block diagram

This block diagram shows you all the provided functions of the library. Three groups of functions are available:

- Initialize/shutdown: Starts and stops the ChipLib.
- Configuration: Allows the user to configure the ChipLib.
- Measurement: Handles the measurements.

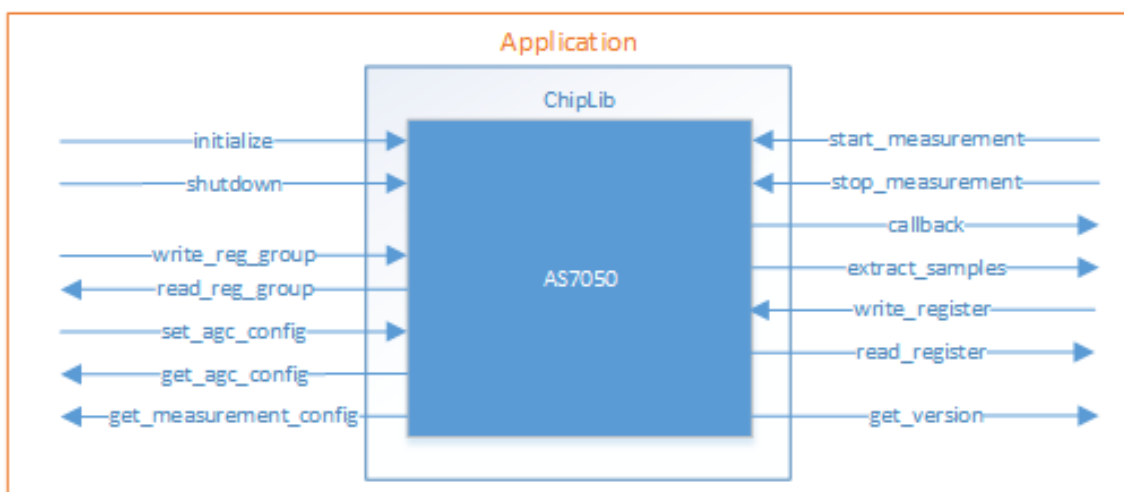


Figure 2 ChipLib API

The functions and their arguments are described in section [ChipLib Functions](#).

3 OSAL

The following pages describe the OSAL (operating system abstraction layer) of the ChipLib. This is used to encapsulate platform- and operating system- specific function to a separate layer, where anybody can customize this interface to their requirements.

This interface is implemented as simply as possible. No complex data types are used and dependencies between functions are as small as possible. An overview of the OSAL is shown in the next section.

3.1 Block diagram

The following figure describes the dependencies of ChipLib. Inside the OSAL, functions like I2C and Interrupt handling must be implemented.

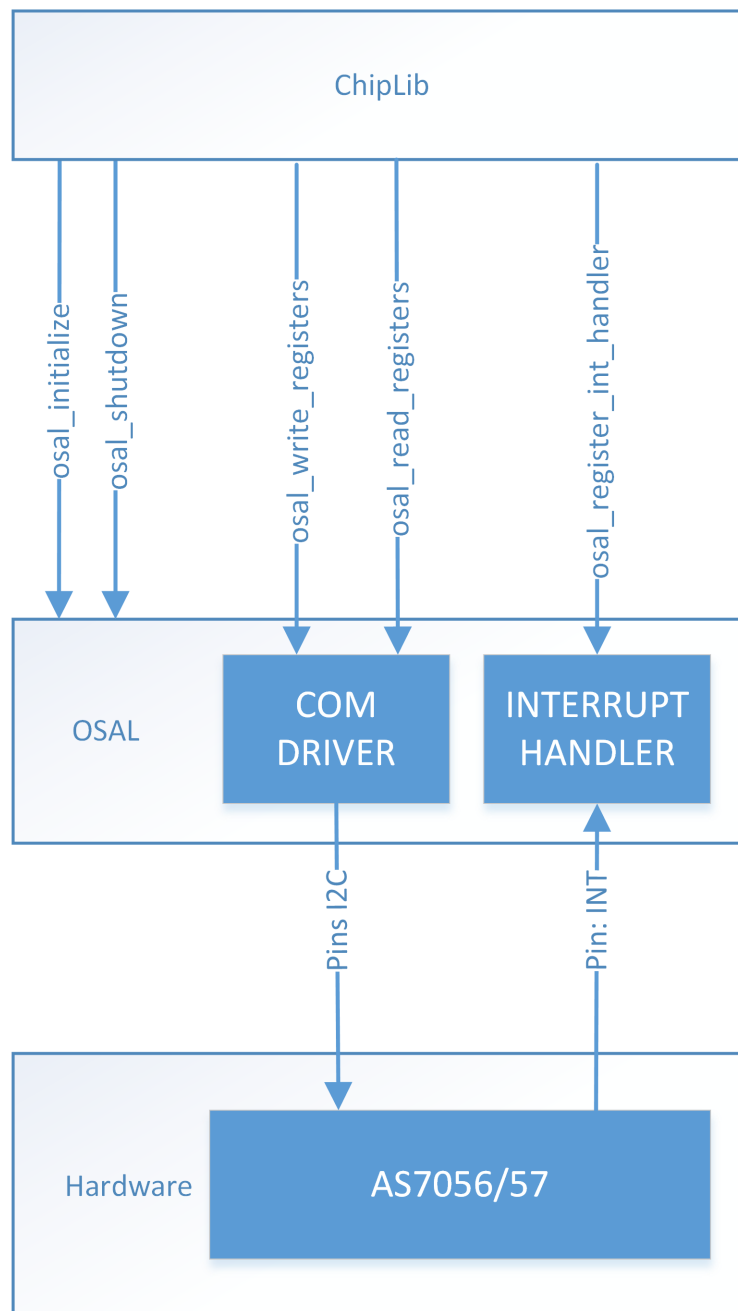


Figure 3 Overview OSAL

Function blocks:

- **COM DRIVER:** Is used to communicate with the sensor via I2C.
- **INTERRUPT HANDLER:** This component handles the interrupt pin and fires an event on falling edge.

3.2 Function dependencies ChipLib - OSAL

The following table describes which OSAL functions are called by which ChipLib functions:

Table 1 Function dependencies ChipLib - OSAL

OSAL functions	ChipLib functions
as7056_57_osal_initialize	as7056_57_initialize
as7056_57_osal_shutdown	as7056_57_shutdown
as7056_57_osal_write_registers , as7056_57_osal_read_registers	as7056_57_initialize , as7056_57_shutdown , as7056_57_write_register , as7056_57_read_register , as7056_57_start_measurement , as7056_57_stop_measurement , as7056_57_get_measurement_config , as7056_57_set_reg_group , as7056_57_get_reg_group
as7056_57_osal_register_int_handler	as7056_57_initialize

3.3 Interfaces

The functions and their arguments are described in section [OSAL Functions](#).

4 Module Documentation

4.1 ChipLib Functions

This is the chip library for ams vital signs chip AS7056/57.

Functions

- [err_code_t](#) [CHIPLIB_DECLDIR](#) [as7056_57_initialize](#) (const [as7056_57_callback_t](#) p_callback, const void *p_cb_param, const char *p_interface_descr)
Initializes the library and the device.
- [err_code_t](#) [CHIPLIB_DECLDIR](#) [as7056_57_shutdown](#) (void)
Stops all internal actions and power down the device.
- [err_code_t](#) [CHIPLIB_DECLDIR](#) [as7056_57_set_reg_group](#) ([as7056_57_reg_group_ids_t](#) id, const uint8_t *p_data, uint8_t size)
Write a register group.
- [err_code_t](#) [CHIPLIB_DECLDIR](#) [as7056_57_get_reg_group](#) ([as7056_57_reg_group_ids_t](#) id, uint8_t *p_data, uint8_t *p_size)
Read a register group.
- [err_code_t](#) [CHIPLIB_DECLDIR](#) [as7056_57_write_register](#) ([as7056_57_reg_addresses_t](#) reg_addr, uint8_t reg_val)
Write register.
- [err_code_t](#) [CHIPLIB_DECLDIR](#) [as7056_57_read_register](#) ([as7056_57_reg_addresses_t](#) reg_addr, uint8_t *p_reg_val)
Read register.

- `err_code_t` CHIPLIB_DECLDIR `as7056_57_get_measurement_config` (`as7056_57_meas_config_t` *p_meas_config)
Read the actual measurement configuration, which was set by the register groups.
- `err_code_t` CHIPLIB_DECLDIR `as7056_57_set_agc_config` (`const agc_configuration_t` *p_agc_configs, `uint8_t` agc_config_num)
Sets the configuration for auto-gain-control (AGC).
- `err_code_t` CHIPLIB_DECLDIR `as7056_57_get_agc_config` (`agc_configuration_t` *p_agc_configs, `uint8_t` *p_agc_config_num)
Gets the configuration for auto-gain-control (AGC).
- `err_code_t` CHIPLIB_DECLDIR `as7056_57_start_measurement` (`void`)
Starts a measurement.
- `err_code_t` CHIPLIB_DECLDIR `as7056_57_stop_measurement` (`void`)
Stops a measurement.
- `err_code_t` CHIPLIB_DECLDIR `as7056_57_get_version` (`as7056_57_version_t` *p_version)
Requests the version information.
- `err_code_t` CHIPLIB_DECLDIR `as7056_57_get_chip_variant` (`uint8_t` size, `char` *p_chip_variant)
Reads the current chip variant.

4.1.1 Detailed Description

This is the chip library for ams vital signs chip AS7056/57.

4.1.2 Function Documentation

4.1.2.1 as7056_57_initialize() `err_code_t` CHIPLIB_DECLDIR `as7056_57_initialize` (
`const as7056_57_callback_t` p_callback,
`const void` * p_cb_param,
`const char` * p_interface_descr)

Initializes the library and the device.

Following tasks will be done here:

- Initialize hardware abstraction layer.
- Reset chip.

Note

This function must be called at first, otherwise all other functions return with error code.

Parameters

in	<i>p_callback</i>	Pointer to the callback function, see <code>as7056_57_callback_t</code> .
in	<i>p_cb_param</i>	Optional pointer to an application parameter, which will be transmitted with every callback.
in	<i>p_interface_descr</i>	Chiplib forwards this interface description to <code>as7056_57_osal_initialize</code> .

Return values

<code>ERR_SUCCESS</code>	Function returns without error.
<code>ERR_ARGUMENT</code>	An argument is invalid.
<code>ERR_IDENTIFICATION</code>	The specified sensor was not found.
<code>ERR_DATA_TRANSFER</code>	Error during communication with sensor.

4.1.2.2 as7056_57_shutdown() `err_code_t` CHIPLIB_DECLDIR `as7056_57_shutdown` (
void)

Stops all internal actions and power down the device.

Following tasks will be done here:

- Stops measurement, if running.
- Power down the sensor device.
- Shutdown the hardware abstraction layer.
- Block calling of all other functions, but initialize.

Return values

<code>ERR_SUCCESS</code>	Function returns without error.
<code>ERR_DATA_TRANSFER</code>	Error during communication with sensor.

4.1.2.3 as7056_57_set_reg_group() `err_code_t` CHIPLIB_DECLDIR `as7056_57_set_reg_group` (
`as7056_57_reg_group_ids_t` *id*,
const uint8_t * *p_data*,
uint8_t *size*)

Write a register group.

This function configures the sensor by directly writing values to a group of sensor registers.

Parameters

in	<i>id</i>	Identification number of the register group, see as7056_57_reg_group_ids .
in	<i>p_data</i>	Pointer to the register data, like as7056_57_config_led_t .
in	<i>size</i>	Sets the size of register data in bytes.

Return values

ERR_SUCCESS	Function returns without error.
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize first.
ERR_ARGUMENT	Register group id is not supported.
ERR_POINTER	Detected NULL pointer for <i>p_data</i> .
ERR_SIZE	Size of the data buffer is wrong.
ERR_DATA_TRANSFER	Error during communication with sensor.

4.1.2.4 as7056_57_get_reg_group() `err_code_t` CHIPLIB_DECLDIR as7056_57_get_reg_group (
[as7056_57_reg_group_ids_t](#) *id*,
uint8_t * *p_data*,
uint8_t * *p_size*)

Read a register group.

Reads the actual register data into a register group structure.

Parameters

in	<i>id</i>	Identification number of the register group, see as7056_57_reg_group_ids .
out	<i>p_data</i>	Pointer, where the data of the register group can be saved.
in, out	<i>p_size</i>	IN: Maximum buffer size, OUT: Size in byte of the register group data. The maximum size is defined in AS7056_57_MAX_GROUP_SIZE .

Return values

ERR_SUCCESS	Function returns without error.
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize at first.
ERR_ARGUMENT	Register group id is not supported.
ERR_POINTER	Detected NULL pointer on arguments.
ERR_SIZE	Size of the data buffer is wrong.
ERR_DATA_TRANSFER	Error during communication with sensor.

4.1.2.5 as7056_57_write_register() `err_code_t` CHIPLIB_DECLDIR as7056_57_write_register (

```
as7056_57_reg_addresses_t reg_addr,
uint8_t reg_val )
```

Write register.

This function sets the value of a single sensor register.

Parameters

in	<i>reg_addr</i>	Register address of the sensor, see as7056_57_reg_addresses .
in	<i>reg_val</i>	New register value, which shall be written to the register address.

Return values

ERR_SUCCESS	Function returns without error.
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize at first.
ERR_DATA_TRANSFER	Error during communication with sensor.

4.1.2.6 as7056_57_read_register() `err_code_t` CHIPLIB_DECLDIR as7056_57_read_register (
[as7056_57_reg_addresses_t](#) *reg_addr*,
uint8_t * *p_reg_val*)

Read register.

This function gets the value of a single sensor register.

Parameters

in	<i>reg_addr</i>	Register address of the sensor, see as7056_57_reg_addresses .
out	<i>p_reg_val</i>	Actual register value, which was read from the register address.

Return values

ERR_SUCCESS	Function returns without error.
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize at first.
ERR_POINTER	Detected NULL pointer for <i>p_reg_val</i> .
ERR_DATA_TRANSFER	Error during communication with sensor.

4.1.2.7 as7056_57_get_measurement_config() `err_code_t` CHIPLIB_DECLDIR as7056_57_get_measurement←
_config (
[as7056_57_meas_config_t](#) * *p_meas_config*)

Read the actual measurement configuration, which was set by the register groups.

Note

The measurement configuration can be changed after set of new register blocks or writing to single registers. Therefore, read it back after finished configuration to get the actual values.

Parameters

out	<i>p_meas_config</i>	Actual measurement configuration, See as7056_57_meas_config_t
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Return values

ERR_SUCCESS	Function returns without error.
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize at first.
ERR_POINTER	Detected NULL pointer for p_meas_config.
ERR_DATA_TRANSFER	Error during communication with sensor.

4.1.2.8 as7056_57_set_agc_config() `err_code_t` CHIPLIB_DECLDIR as7056_57_set_agc_config (
const agc_configuration_t * p_agc_configs,
uint8_t agc_config_num)

Sets the configuration for auto-gain-control (AGC).

Parameters

in	<i>p_agc_configs</i>	Pointer to AGC configuration information for up to 4 channels.
in	<i>agc_config_num</i>	Number of AGC channels.

Return values

ERR_SUCCESS	Function returns without error.
ERR_ARGUMENT	At least one parameter is wrong.
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize at first.
ERR_POINTER	Detected NULL pointer for p_agc_config.

4.1.2.9 as7056_57_get_agc_config() `err_code_t` CHIPLIB_DECLDIR as7056_57_get_agc_config (
agc_configuration_t * p_agc_configs,
uint8_t * p_agc_config_num)

Gets the configuration for auto-gain-control (AGC).

Parameters

out	<i>p_agc_configs</i>	Pointer to store AGC configuration for activated channels.
in, out	<i>p_agc_config_num</i>	Number of channels. IN: Defines how many AGC channel configurations can be saved inside p_agc_configs. OUT: number of enabled AGC channels.

Return values

<i>ERR_SUCCESS</i>	Function returns without error.
<i>ERR_PERMISSION</i>	Access to the library is blocked, call as7056_57_initialize at first.
<i>ERR_POINTER</i>	Detected NULL pointer for p_agc_config.
<i>ERR_SIZE</i>	Value pointed to by p_agc_config_num is too small.

4.1.2.10 as7056_57_start_measurement() [*err_code_t*](#) `CHIPLIB_DECLDIR as7056_57_start_measurement`
(
 void)

Starts a measurement.

Return values

<i>ERR_SUCCESS</i>	Function returns without error.
<i>ERR_PERMISSION</i>	Access to the library is blocked, call as7056_57_initialize at first.
<i>ERR_DATA_TRANSFER</i>	Error during communication with sensor.
<i>ERR_CONFIG</i>	No sub-sample is enabled, AOC and AGC are enabled in parallel.

4.1.2.11 as7056_57_stop_measurement() [*err_code_t*](#) `CHIPLIB_DECLDIR as7056_57_stop_measurement` (
 void)

Stops a measurement.

Return values

<i>ERR_SUCCESS</i>	Function returns without error.
<i>ERR_PERMISSION</i>	Access to the library is blocked, call as7056_57_initialize at first.
<i>ERR_DATA_TRANSFER</i>	Error during communication with sensor.

4.1.2.12 as7056_57_get_version() `err_code_t` CHIPLIB_DECLDIR as7056_57_get_version (
 `as7056_57_version_t * p_version`)

Requests the version information.

Parameters

out	<code>p_version</code>	Pointer to memory, where the version information can be saved. See as7056_57_version_t .
-----	------------------------	--

Return values

ERR_SUCCESS	Function returns without error.
ERR_POINTER	Detected NULL pointer for <code>p_version</code> .

4.1.2.13 as7056_57_get_chip_variant() `err_code_t` CHIPLIB_DECLDIR as7056_57_get_chip_variant (
 `uint8_t size,`
 `char * p_chip_variant`)

Reads the current chip variant.

Parameters

in	<code>size</code>	Maximum size of <code>p_chip_variant</code> . At least 8 bytes.
out	<code>p_chip_variant</code>	Memory where the name of the connected chip will be saved: 'AS7056', 'AS7057', 'unknown'. The string is NULL-terminated.

Return values

ERR_SUCCESS	Function returns without error.
ERR_POINTER	Detected NULL pointer for <code>p_version</code> .
ERR_PERMISSION	Access to the library is blocked, call as7056_57_initialize at first.
ERR_DATA_TRANSFER	Error during communication with sensor.
ERR_SIZE	Size is not large enough.

4.2 OSAL Functions

This is the abstraction layer for the chip library.

Typedefs

- typedef [err_code_t](#)(* [as7056_57_osal_interrupt_t](#)) (void)
Callback function which will be called if a new interrupt notification is available.

Functions

- [err_code_t](#) [as7056_57_osal_initialize](#) (const char *p_interface_desc)
Initialization of the hardware abstraction layer.
- [err_code_t](#) [as7056_57_osal_shutdown](#) (void)
Shutdown of the hardware abstraction layer.
- [err_code_t](#) [as7056_57_osal_write_registers](#) (uint8_t address, uint16_t number, const uint8_t *p_values)
Set register values inside the chip.
- [err_code_t](#) [as7056_57_osal_read_registers](#) (uint8_t address, uint16_t number, uint8_t *p_values)
Get register values from the chip.
- [err_code_t](#) [as7056_57_osal_register_int_handler](#) (const [as7056_57_osal_interrupt_t](#) callback_function)
Registers and unregisters an interrupt handler.

4.2.1 Detailed Description

This is the abstraction layer for the chip library.

Some functions of the chip library are dependent on the operating system. These functions concern initialization, shutdown, I2C transfers, and interrupt handling. They need to be implemented specifically for each application.

4.2.2 Typedef Documentation

4.2.2.1 [as7056_57_osal_interrupt_t](#) typedef [err_code_t](#)(* [as7056_57_osal_interrupt_t](#)) (void)

Callback function which will be called if a new interrupt notification is available.

This callback function is registered using the function [as7056_57_osal_register_int_handler](#).

4.2.3 Function Documentation

4.2.3.1 as7056_57_osal_initialize() `err_code_t as7056_57_osal_initialize (`
`const char * p_interface_desc )`

Initialization of the hardware abstraction layer.

- Initialization of global parameters.
- Opens the interface to the sensor.

Note

This function must be called first!

Parameters

in	<code>p_interface_desc</code>	Can be used to transfer special initialization data like interface description.
----	-------------------------------	---

Return values

<code>ERR_SUCCESS</code>	Function returns without error.
<code>ERR_ARGUMENT</code>	Argument content is not supported.
<code>ERR_COM_INTERFACE</code>	The interface to the sensor is faulty.

4.2.3.2 as7056_57_osal_shutdown() `err_code_t as7056_57_osal_shutdown (`
`void )`

Shutdown of the hardware abstraction layer.

- Closes the interface to the sensor
- Deactivates the ENABLE-Pin

Note

This function must be called for cleanup.

Return values

<code>ERR_SUCCESS</code>	Function returns without error.
<code>ERR_COM_INTERFACE</code>	The interface to the sensor is faulty.

4.2.3.3 as7056_57_osal_write_registers() `err_code_t as7056_57_osal_write_registers (`
`uint8_t address,`
`uint16_t number,`
`const uint8_t * p_values )`

Set register values inside the chip.

Parameters

in	<i>address</i>	Register address of the first register to write.
in	<i>number</i>	Number of registers to write. The register address is auto-incremented by the sensor.
in	<i>p_values</i>	Pointer to register values.

Return values

<i>ERR_SUCCESS</i>	Function returns without error.
<i>ERR_POINTER</i>	NULL pointer detected.
<i>ERR_DATA_TRANSFER</i>	Data transfer error, like bus error or timeout.
<i>ERR_PERMISSION</i>	Library was not initialized using as7056_57_osal_initialize .

4.2.3.4 as7056_57_osal_read_registers() `err_code_t as7056_57_osal_read_registers (`
`uint8_t address,`
`uint16_t number,`
`uint8_t * p_values )`

Get register values from the chip.

Parameters

in	<i>address</i>	Register address of the first register to read.
in	<i>number</i>	Number of registers to read. The register address is auto-incremented by the sensor.
out	<i>p_values</i>	Pointer to buffer where the read register values are written to.

Return values

<i>ERR_SUCCESS</i>	Function returns without error.
<i>ERR_POINTER</i>	NULL pointer detected.
<i>ERR_DATA_TRANSFER</i>	Data transfer error, like bus error or timeout.
<i>ERR_PERMISSION</i>	Library was not initialized using as7056_57_osal_initialize .

4.2.3.5 as7056_57_osal_register_int_handler() `err_code_t as7056_57_osal_register_int_handler (`
`const as7056_57_osal_interrupt_t callback_function )`

Registers and unregisters an interrupt handler.

If the function pointer is NULL, then the old callback will be unregistered.

Parameters

in	<i>callback_function</i>	Pointer to callback function or NULL.
----	--------------------------	---------------------------------------

Return values

<i>ERR_SUCCESS</i>	Function returns without error.
<i>ERR_PERMISSION</i>	Library was not initialized using as7056_57_osal_initialize .

4.3 Error Codes

Typedefs

- typedef enum `error_codes` `err_code_t`

Enumerations

- enum `error_codes` {
 `ERR_SUCCESS` = 0,
 `ERR_PERMISSION` = 1,
 `ERR_MESSAGE` = 2,
 `ERR_MESSAGE_SIZE` = 3,
 `ERR_POINTER` = 4,
 `ERR_ACCESS` = 5,
 `ERR_ARGUMENT` = 6,
 `ERR_SIZE` = 7,
 `ERR_NOT_SUPPORTED` = 8,
 `ERR_TIMEOUT` = 9,
 `ERR_CHECKSUM` = 10,
 `ERR_OVERFLOW` = 11,
 `ERR_EVENT` = 12,
 `ERR_INTERRUPT` = 13,
 `ERR_TIMER_ACCESS` = 14,
 `ERR_LED_ACCESS` = 15,
 `ERR_TEMP_SENSOR_ACCESS` = 16,
 `ERR_DATA_TRANSFER` = 17,
 `ERR_FIFO` = 18,
 `ERR_OVER_TEMP` = 19,
 `ERR_IDENTIFICATION` = 20,
 `ERR_COM_INTERFACE` = 21,
 `ERR_SYNCHRONISATION` = 22,
 `ERR_PROTOCOL` = 23,
 `ERR_MEMORY` = 24,
 `ERR_THREAD` = 25,
 `ERR_SPI` = 26,
 `ERR_DAC_ACCESS` = 27,
 `ERR_I2C` = 28,
 `ERR_NO_DATA` = 29,
 `ERR_SYSTEM_CONFIG` = 30,
 `ERR_USB_ACCESS` = 31,
 `ERR_ADC_ACCESS` = 32,
 `ERR_SENSOR_CONFIG` = 33,
 `ERR SATURATION` = 34,
 `ERR_MUTEX` = 35,
 `ERR_ACCELEROMETER` = 36,
 `ERR_CONFIG` = 37,
 `ERR_BLE` = 38,
 `ERR_FILE` = 39,
 `ERR_DATA` = 40,
 `ERR_BUSY` = 41 }

4.3.1 Detailed Description

Generic error codes used by ams libraries.

4.3.2 Typedef Documentation

4.3.2.1 `err_code_t` typedef enum `error_codes` `err_code_t`

This definition will be used for function return values.

4.3.3 Enumeration Type Documentation

4.3.3.1 `error_codes` enum `error_codes`

Error Codes.

Enumerator

<code>ERR_SUCCESS</code>	Operation was successful.
<code>ERR_PERMISSION</code>	Operation is not permitted.
<code>ERR_MESSAGE</code>	Message is invalid. (Unsupported message type, incorrect CRC, ...)
<code>ERR_MESSAGE_SIZE</code>	Message has the wrong size.
<code>ERR_POINTER</code>	Pointer is invalid. (NULL pointer, pointer wrong memory region, ...)
<code>ERR_ACCESS</code>	Access is denied.
<code>ERR_ARGUMENT</code>	Argument is invalid.
<code>ERR_SIZE</code>	An argument has the wrong size.
<code>ERR_NOT_SUPPORTED</code>	Function is not supported or not implemented.
<code>ERR_TIMEOUT</code>	Operation timed out.
<code>ERR_CHECKSUM</code>	Checksum comparison failed.
<code>ERR_OVERFLOW</code>	Data overflow occurred.
<code>ERR_EVENT</code>	Getting or setting an event failed. (Event queue is full or empty, unexpected event received, ...)
<code>ERR_INTERRUPT</code>	Getting or setting an interrupt failed. (Interrupt resource is not available, ...)
<code>ERR_TIMER_ACCESS</code>	Accessing the timer peripheral failed.
<code>ERR_LED_ACCESS</code>	Accessing the LED peripheral failed.
<code>ERR_TEMP_SENSOR_ACCESS</code>	Accessing the temperature sensor failed.
<code>ERR_DATA_TRANSFER</code>	Communication error occurred.
<code>ERR_FIFO</code>	FIFO operation failed.
<code>ERR_OVER_TEMP</code>	Overtemperature detected.

Enumerator

ERR_IDENTIFICATION	Sensor identification failed.
ERR_COM_INTERFACE	Generic communication interface error. (Communication interface is not available, error while opening or closing a communication interface, ...)
ERR_SYNCHRONISATION	Synchronization error occurred.
ERR_PROTOCOL	Generic protocol error occurred.
ERR_MEMORY	Memory allocation error occurred.
ERR_THREAD	Thread handling operation failed.
ERR_SPI	Accessing the SPI peripheral failed.
ERR_DAC_ACCESS	Accessing the DAC peripheral failed.
ERR_I2C	Accessing the I2C peripheral failed.
ERR_NO_DATA	No data available.
ERR_SYSTEM_CONFIG	System configuration failed. (System resource is not available, system resource generates an error, ...)
ERR_USB_ACCESS	Accessing the USB peripheral failed.
ERR_ADC_ACCESS	Accessing the ADC peripheral failed.
ERR_SENSOR_CONFIG	Sensor configuration failed.
ERR_SATURATION	Saturation detected.
ERR_MUTEX	Mutex handling operation failed.
ERR_ACCELEROMETER	Accessing the accelerometer failed.
ERR_CONFIG	Software component is unusable due to incomplete or incorrect configuration.
ERR_BLE	BLE stack handling operation failed.
ERR_FILE	File handling operation failed.
ERR_DATA	Internal data inconsistency detected.
ERR_BUSY	Module is busy.

4.4 ChipLib Definitions

Description of the used data types.

Data Structures

- struct [as7056_57_version](#)
- struct [as7056_57_interrupt_enable_t](#)
- union [as7056_57_config_aoc_t](#)
- union [as7056_57_config_led_t](#)
- union [as7056_57_config_pd_t](#)
- union [as7056_57_config_mod_t](#)
- union [as7056_57_config_sinc_t](#)
- union [as7056_57_config_seq_t](#)
- union [as7056_57_config_pp_t](#)
- union [as7056_57_config_ref_t](#)
- union [as7056_57_config_ctrl_t](#)
- union [as7056_57_config_standby_t](#)
- union [as7056_57_config_fifo_t](#)
- union [as7056_57_config_prox_t](#)
- struct [as7056_57_meas_config_t](#)

Macros

- #define [AS7056_57_SILICON_ID](#) 0x8A
- #define [AS7056_57_MAX_GROUP_SIZE](#) 30
- #define [AS7056_57_FIFO_SAMPLE_SIZE](#) 3
- #define [AS7056_57_FIFO_DATA_BUFFER_SIZE](#) 1536
- #define [M_SUB_SAMPLE_ID_TO_FLAG](#)(sub_sample_id) ([as7056_57_sub_sample_flags_t](#))(1 << (sub_↵
sample_id - 1))

Typedefs

- typedef uint8_t [as7056_57_reg_group_ids_t](#)
- typedef uint32_t [as7056_57_sub_sample_flags_t](#)
- typedef uint8_t [as7056_57_sub_sample_ids_t](#)
- typedef uint8_t [as7056_57_seq_ids_t](#)
- typedef uint8_t [as7056_57_reg_addresses_t](#)
- typedef struct [as7056_57_version](#) [as7056_57_version_t](#)
- typedef void(* [as7056_57_callback_t](#)) ([err_code_t](#) error, uint8_t *p_fifo_data, uint16_t fifo_data_size, agc↵
_status_t *p_agc_statuses, uint8_t agc_statuses_num, uint32_t chip_events, const void *p_cb_param)

Callback function, which transfers the measurement results to the application.

Enumerations

- enum `as7056_57_reg_group_ids` {
`AS7056_57_REG_GROUP_ID_CTRL` = 0,
`AS7056_57_REG_GROUP_ID_REF` = 1,
`AS7056_57_REG_GROUP_ID_STBY` = 2,
`AS7056_57_REG_GROUP_ID_PROX` = 3,
`AS7056_57_REG_GROUP_ID_SEQ` = 4,
`AS7056_57_REG_GROUP_ID_LED` = 5,
`AS7056_57_REG_GROUP_ID_PD` = 6,
`AS7056_57_REG_GROUP_ID_AOC` = 7,
`AS7056_57_REG_GROUP_ID_MOD` = 8,
`AS7056_57_REG_GROUP_ID_SINC` = 9,
`AS7056_57_REG_GROUP_ID_PP` = 10,
`AS7056_57_REG_GROUP_ID_FIFO` = 11,
`AS7056_57_REG_GROUP_ID_NUM` = 12 }
- enum `as7056_57_sub_sample_flags` {
`AS7056_57_SUB_SAMPLE_FLAG_NONE` = 0x00000000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB1` = 0x00000001,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB2` = 0x00000002,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB3` = 0x00000004,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB4` = 0x00000008,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB5` = 0x00000010,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB6` = 0x00000020,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB7` = 0x00000040,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB8` = 0x00000080,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB1` = 0x00000100,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB2` = 0x00000200,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB3` = 0x00000400,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB4` = 0x00000800,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB5` = 0x00001000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB6` = 0x00002000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB7` = 0x00004000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB8` = 0x00008000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB1` = 0x00010000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB2` = 0x00020000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB3` = 0x00040000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB4` = 0x00080000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB1` = 0x00100000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB2` = 0x00200000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB3` = 0x00400000,
`AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB4` = 0x00800000,
`AS7056_57_SUB_SAMPLE_FLAG_SAR` = 0x01000000,
`AS7056_57_SUB_SAMPLE_FLAG_AOC` = 0x02000000,
`AS7056_57_SUB_SAMPLE_FLAG_SEQ1` = 0x0000FFFF,
`AS7056_57_SUB_SAMPLE_FLAG_SEQ2` = 0x00FF0000 }
- enum `as7056_57_sub_sample_ids` {
`AS7056_57_SUB_SAMPLE_ID_DISABLED` = 0,
`AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB1` = 1,
`AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB2` = 2,
`AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB3` = 3,
`AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB4` = 4,
`AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB5` = 5,
`AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB6` = 6,

```

AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB7 = 7,
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB8 = 8,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB1 = 9,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB2 = 10,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB3 = 11,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB4 = 12,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB5 = 13,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB6 = 14,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB7 = 15,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB8 = 16,
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB1 = 17,
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB2 = 18,
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB3 = 19,
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB4 = 20,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB1 = 21,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB2 = 22,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB3 = 23,
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB4 = 24,
AS7056_57_SUB_SAMPLE_ID_MAX = 24 }

• enum as7056_57_fifo_data_marker {
    AS7056_57_FIFO_DATA_MARKER_MOD1_SEQ1_FIRST = 0,
    AS7056_57_FIFO_DATA_MARKER_MOD1_SEQ2_FIRST = 1,
    AS7056_57_FIFO_DATA_MARKER_MOD1_OTHER = 2,
    AS7056_57_FIFO_DATA_MARKER_MOD2_SEQ1_FIRST = 3,
    AS7056_57_FIFO_DATA_MARKER_MOD2_SEQ2_FIRST = 4,
    AS7056_57_FIFO_DATA_MARKER_MOD2_OTHER = 5,
    AS7056_57_FIFO_DATA_MARKER_SAR = 6,
    AS7056_57_FIFO_DATA_MARKER_AOC = 7 }

• enum as7056_57_seq_ids {
    AS7056_57_ID_SEQ_NONE = 0,
    AS7056_57_ID_SEQ1 = 1,
    AS7056_57_ID_SEQ2 = 2 }

• enum as7056_57_reg_addresses {
    AS7056_57_REGADDR_CONTROL = 0x10,
    AS7056_57_REGADDR_CGB_CFG = 0x11,
    AS7056_57_REGADDR_INT_CFG = 0x12,
    AS7056_57_REGADDR_CSXN_CFG = 0x13,
    AS7056_57_REGADDR_IO_CFG = 0x14,
    AS7056_57_REGADDR_REF_CFGA = 0x15,
    AS7056_57_REGADDR_REF_CFGB = 0x16,
    AS7056_57_REGADDR_MOD_CFGA = 0x19,
    AS7056_57_REGADDR_MOD_CFGB = 0x1A,
    AS7056_57_REGADDR_MOD1_CFGA = 0x1B,
    AS7056_57_REGADDR_MOD1_CFGB = 0x1C,
    AS7056_57_REGADDR_MOD1_CFGC = 0x1D,
    AS7056_57_REGADDR_MOD1_CFGD = 0x1E,
    AS7056_57_REGADDR_MOD1_CFGE = 0x1F,
    AS7056_57_REGADDR_MOD1_CFGF = 0x20,
    AS7056_57_REGADDR_MOD2_CFGA = 0x21,
    AS7056_57_REGADDR_MOD2_CFGB = 0x22,
    AS7056_57_REGADDR_MOD2_CFGC = 0x23,
    AS7056_57_REGADDR_MOD2_CFGD = 0x24,
    AS7056_57_REGADDR_MOD2_CFGE = 0x25,
    AS7056_57_REGADDR_MOD2_CFGF = 0x26,

```

```

AS7056_57_REGADDR_VCSEL_CFG = 0x28,
AS7056_57_REGADDR_SEQ1_LED1_CURR = 0x29,
AS7056_57_REGADDR_SEQ2_LED1_CURR = 0x2A,
AS7056_57_REGADDR_SEQ1_LED2_CURR = 0x2B,
AS7056_57_REGADDR_SEQ2_LED2_CURR = 0x2C,
AS7056_57_REGADDR_SEQ1_LED3_CURR = 0x2D,
AS7056_57_REGADDR_SEQ2_LED3_CURR = 0x2E,
AS7056_57_REGADDR_LED_SEQ1_SUB12 = 0x2F,
AS7056_57_REGADDR_LED_SEQ1_SUB34 = 0x30,
AS7056_57_REGADDR_LED_SEQ1_SUB56 = 0x31,
AS7056_57_REGADDR_LED_SEQ1_SUB78 = 0x32,
AS7056_57_REGADDR_LED_SEQ2_SUB12 = 0x33,
AS7056_57_REGADDR_LED_SEQ2_SUB34 = 0x34,
AS7056_57_REGADDR_LED_LOWVDS_WAIT = 0x35,
AS7056_57_REGADDR_PP_CFG = 0x37,
AS7056_57_REGADDR_SEQ1_SUB12_PP = 0x38,
AS7056_57_REGADDR_SEQ1_SUB34_PP = 0x39,
AS7056_57_REGADDR_SEQ1_SUB56_PP = 0x3A,
AS7056_57_REGADDR_SEQ1_SUB78_PP = 0x3B,
AS7056_57_REGADDR_SEQ2_SUB12_PP = 0x3C,
AS7056_57_REGADDR_SEQ2_SUB34_PP = 0x3D,
AS7056_57_REGADDR_IRQ_ENABLE = 0x3F,
AS7056_57_REGADDR_SEQ_SAMPLE = 0x40,
AS7056_57_REGADDR_SEQ_SUB_WAIT = 0x41,
AS7056_57_REGADDR_SEQ_MODCONF = 0x42,
AS7056_57_REGADDR_SEQ_CONFIG = 0x43,
AS7056_57_REGADDR_SEQ_SAR_WAIT = 0x44,
AS7056_57_REGADDR_SEQ_LED_INIT = 0x45,
AS7056_57_REGADDR_SEQ_FREQ_L = 0x46,
AS7056_57_REGADDR_SEQ_FREQ_H = 0x47,
AS7056_57_REGADDR_SEQ1_FREQDIV_L = 0x48,
AS7056_57_REGADDR_SEQ1_FREQDIV_H = 0x49,
AS7056_57_REGADDR_SEQ2_FREQDIV_L = 0x4A,
AS7056_57_REGADDR_SEQ2_FREQDIV_H = 0x4B,
AS7056_57_REGADDR_MOD1_SEQ1_SUB_EN = 0x4C,
AS7056_57_REGADDR_MOD1_SEQ2_SUB_EN = 0x4D,
AS7056_57_REGADDR_MOD2_SEQ1_SUB_EN = 0x4E,
AS7056_57_REGADDR_MOD2_SEQ2_SUB_EN = 0x4F,
AS7056_57_REGADDR_SEQ1_MODE_A = 0x50,
AS7056_57_REGADDR_SEQ1_MODE_B = 0x51,
AS7056_57_REGADDR_SEQ2_MODE = 0x52,
AS7056_57_REGADDR_PD_SEQ1_SUB1 = 0x53,
AS7056_57_REGADDR_PD_SEQ1_SUB2 = 0x54,
AS7056_57_REGADDR_PD_SEQ1_SUB3 = 0x55,
AS7056_57_REGADDR_PD_SEQ1_SUB4 = 0x56,
AS7056_57_REGADDR_PD_SEQ1_SUB5 = 0x57,
AS7056_57_REGADDR_PD_SEQ1_SUB6 = 0x58,
AS7056_57_REGADDR_PD_SEQ1_SUB7 = 0x59,
AS7056_57_REGADDR_PD_SEQ1_SUB8 = 0x5A,
AS7056_57_REGADDR_PD_SEQ2_SUB1 = 0x5B,
AS7056_57_REGADDR_PD_SEQ2_SUB2 = 0x5C,
AS7056_57_REGADDR_PD_SEQ2_SUB3 = 0x5D,
AS7056_57_REGADDR_PD_SEQ2_SUB4 = 0x5E,
AS7056_57_REGADDR_PDSEL_CFG = 0x5F,

```

```

AS7056_57_REGADDR_SEQ1_SINC_CFGA = 0x61,
AS7056_57_REGADDR_SEQ1_SINC_CFGB = 0x62,
AS7056_57_REGADDR_SEQ1_SINC_CFGC = 0x63,
AS7056_57_REGADDR_SEQ2_SINC_CFGA = 0x64,
AS7056_57_REGADDR_SEQ2_SINC_CFGB = 0x65,
AS7056_57_REGADDR_SEQ2_SINC_CFGC = 0x66,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB1 = 0x70,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB2 = 0x71,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB3 = 0x72,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB4 = 0x73,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB5 = 0x74,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB6 = 0x75,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB7 = 0x76,
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB8 = 0x77,
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB1 = 0x78,
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB2 = 0x79,
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB3 = 0x7A,
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB4 = 0x7B,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB1 = 0x7C,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB2 = 0x7D,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB3 = 0x7E,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB4 = 0x7F,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB5 = 0x80,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB6 = 0x81,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB7 = 0x82,
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB8 = 0x83,
AS7056_57_REGADDR_AOC_LED OFF = 0x84,
AS7056_57_REGADDR_AOC_CFG = 0x85,
AS7056_57_REGADDR_AOC_MOD1_THH = 0x86,
AS7056_57_REGADDR_AOC_MOD1_THL = 0x87,
AS7056_57_REGADDR_AOC_MOD2_THH = 0x88,
AS7056_57_REGADDR_AOC_MOD2_THL = 0x89,
AS7056_57_REGADDR_AOC_SAR_THRES = 0x8A,
AS7056_57_REGADDR_MOD1_SEQ1_AOC_EN = 0x8B,
AS7056_57_REGADDR_MOD2_SEQ1_AOC_EN = 0x8C,
AS7056_57_REGADDR_PROX_CFG = 0x98,
AS7056_57_REGADDR_PROX_OVS = 0x99,
AS7056_57_REGADDR_PROX_THH_L = 0x9A,
AS7056_57_REGADDR_PROX_THH_H = 0x9B,
AS7056_57_REGADDR_PROX_THL_L = 0x9C,
AS7056_57_REGADDR_PROX_THL_H = 0x9D,
AS7056_57_REGADDR_STANDBY_ON = 0xA0,
AS7056_57_REGADDR_STANDBY_EN1 = 0xA1,
AS7056_57_REGADDR_STANDBY_EN2 = 0xA2,
AS7056_57_REGADDR_STANDBY_EN3 = 0xA3,
AS7056_57_REGADDR_STANDBY_EN4 = 0xA4,
AS7056_57_REGADDR_STANDBY_EN5 = 0xA5,
AS7056_57_REGADDR_STANDBY_EN6 = 0xA6,
AS7056_57_REGADDR_STANDBY_EN7 = 0xA7,
AS7056_57_REGADDR_FIFO_THRESHOLD = 0xD0,
AS7056_57_REGADDR_FIFO_CTRL = 0xD1,
AS7056_57_REGADDR_PRODUCT_ID = 0xEC,
AS7056_57_REGADDR_SILICON_ID = 0xED,
AS7056_57_REGADDR_REVISION = 0xEE,

```

```

AS7056_57_REGADDR_CHIP_CTRL = 0xEF,
AS7056_57_REGADDR_SEQ_START = 0xF0,
AS7056_57_REGADDR_STATUS_CGBA = 0xF3,
AS7056_57_REGADDR_STATUS_CGBB = 0xF4,
AS7056_57_REGADDR_STATUS_SEQ = 0xF5,
AS7056_57_REGADDR_STATUS_LED = 0xF6,
AS7056_57_REGADDR_STATUS_ASAT = 0xF7,
AS7056_57_REGADDR_STATUS_VCSEL = 0xF8,
AS7056_57_REGADDR_STATUS_PROX = 0xF9,
AS7056_57_REGADDR_STATUS = 0xFA,
AS7056_57_REGADDR_FIFO_LEVEL0 = 0xFB,
AS7056_57_REGADDR_FIFO_LEVEL1 = 0xFC,
AS7056_57_REGADDR_FIFOL = 0xFD,
AS7056_57_REGADDR_FIFOM = 0xFE,
AS7056_57_REGADDR_FIFOH = 0xFF }
• enum as7056_57_chip_events {
AS7056_57_CHIP_EVENT_PROX_LOW_OFF = 0x00000001,
AS7056_57_CHIP_EVENT_PROX_LOW_ON = 0x00000002,
AS7056_57_CHIP_EVENT_PROX_LOW = 0x00000004,
AS7056_57_CHIP_EVENT_PROX_HIGH_OFF = 0x00000010,
AS7056_57_CHIP_EVENT_PROX_HIGH_ON = 0x00000020,
AS7056_57_CHIP_EVENT_PROX_HIGH = 0x00000040,
AS7056_57_CHIP_EVENT_VCSEL_WD = 0x00000100,
AS7056_57_CHIP_EVENT_VCSEL_SHORT_VDD = 0x00000200,
AS7056_57_CHIP_EVENT_VCSEL_SHORT_VSS = 0x00000400,
AS7056_57_CHIP_EVENT_MOD2_ASAT = 0x000F0000,
AS7056_57_CHIP_EVENT_MOD1_ASAT = 0x00F00000,
AS7056_57_CHIP_EVENT_LED_LOWVDS = 0x07000000,
AS7056_57_CHIP_EVENT_SEQ_ERROR = 0x10000000,
AS7056_57_CHIP_EVENT_SEQ_END = 0x20000000,
AS7056_57_CHIP_EVENT_PROX_BITS_SHIFT = 0,
AS7056_57_CHIP_EVENT_VCSEL_BITS_SHIFT = 8,
AS7056_57_CHIP_EVENT_ASAT_BITS_SHIFT = 16,
AS7056_57_CHIP_EVENT_LED_BITS_SHIFT = 24,
AS7056_57_CHIP_EVENT_SEQ_BITS_SHIFT = 28 }

```

4.4.1 Detailed Description

Description of the used data types.

These are the type definitions used by AS7056_57 chip library.

4.4.2 Macro Definition Documentation

4.4.2.1 AS7056_57_SILICON_ID `#define AS7056_57_SILICON_ID 0x8A`

Silicon ID of AS7056/57

4.4.2.2 AS7056_57_MAX_GROUP_SIZE `#define AS7056_57_MAX_GROUP_SIZE 30`

Maximum combined size of the register values of a single register group

4.4.2.3 AS7056_57_FIFO_SAMPLE_SIZE `#define AS7056_57_FIFO_SAMPLE_SIZE 3`

Size of one sample inside the FIFO

4.4.2.4 AS7056_57_FIFO_DATA_BUFFER_SIZE `#define AS7056_57_FIFO_DATA_BUFFER_SIZE 1536`

Size of the buffer which holds the FIFO data.

4.4.2.5 M_SUB_SAMPLE_ID_TO_FLAG `#define M_SUB_SAMPLE_ID_TO_FLAG( sub_sample_id ) (as7056_57_sub_sample_flags_t)(1 << (sub_sample_id - 1))`

Calculate the sub-sample flag from the ID

4.4.3 Typedef Documentation

4.4.3.1 `as7056_57_reg_group_ids_t` `typedef uint8_t as7056_57_reg_group_ids_t`

Type for `as7056_57_reg_group_ids`.

4.4.3.2 `as7056_57_sub_sample_flags_t` `typedef uint32_t as7056_57_sub_sample_flags_t`

Type for `as7056_57_sub_sample_flags`.

4.4.3.3 `as7056_57_sub_sample_ids_t` `typedef uint8_t as7056_57_sub_sample_ids_t`

Type for `as7056_57_sub_sample_ids`

4.4.3.4 `as7056_57_seq_ids_t` `typedef uint8_t as7056_57_seq_ids_t`

Type for `as7056_57_seq_ids`

4.4.3.5 `as7056_57_reg_addresses_t` `typedef uint8_t as7056_57_reg_addresses_t`

Type for `as7056_57_reg_addresses`

4.4.3.6 `as7056_57_version_t` `typedef struct as7056_57_version as7056_57_version_t`

Version information of the library

4.4.3.7 `as7056_57_callback_t` `typedef void(* as7056_57_callback_t) (err_code_t error, uint8_t *p_fifo_data, uint16_t fifo_data_size, agc_status_t *p_agc_statuses, uint8_t agc_statuses_num, uint32_t chip_events, const void *p_cb_param)`

Callback function, which transfers the measurement results to the application.

This callback type will be registered via the function `as7056_57_initialize`. During the measurement, this function periodically transfers the results.

Parameters

in	<i>error</i>	Default ERR_SUCCESS , otherwise an error occurred during measurement and the measurement is stopped. See error_codes
in	<i>p_fifo_data</i>	Pointer to the measurement data, the content depends on configuration.
in	<i>fifo_data_size</i>	Size of the FIFO data buffer in bytes.
in	<i>p_agc_statuses</i>	Pointer to AGC status structures.
in	<i>agc_statuses_num</i>	Number of AGC status elements.
in	<i>chip_events</i>	Combination of all possible chip event flags but FIFO status. See as7056_57_chip_events
in	<i>p_cb_param</i>	Application parameter which was defined during call of as7056_57_initialize .

4.4.4 Enumeration Type Documentation

4.4.4.1 as7056_57_reg_group_ids enum [as7056_57_reg_group_ids](#)

Definition of the register group IDs

Enumerator

AS7056_57_REG_GROUP_ID_CTRL	Group ID of the register group 'Control'. See as7056_57_config_ctrl_t
AS7056_57_REG_GROUP_ID_REF	Group ID of the register group 'Reference'. See as7056_57_config_ref_t
AS7056_57_REG_GROUP_ID_STBY	Group ID of the register group 'Standby'. See as7056_57_config_standby_t
AS7056_57_REG_GROUP_ID_PROX	Group ID of the register group 'Proximity'. See as7056_57_config_prox_t
AS7056_57_REG_GROUP_ID_SEQ	Group ID of the register group 'Sequencer'. See as7056_57_config_seq_t
AS7056_57_REG_GROUP_ID_LED	Group ID of the register group 'LED'. See as7056_57_config_led_t
AS7056_57_REG_GROUP_ID_PD	Group ID of the register group 'Photodiodes'. See as7056_57_config_pd_t
AS7056_57_REG_GROUP_ID_AOC	Group ID of the register group 'Automatic Offset Control'. See as7056_57_config_aoc_t
AS7056_57_REG_GROUP_ID_MOD	Group ID of the register group 'Modulator'. See as7056_57_config_mod_t
AS7056_57_REG_GROUP_ID_SINC	Group ID of the register group 'Sinc filter'. See as7056_57_config_sinc_t
AS7056_57_REG_GROUP_ID_PP	Group ID of the register group 'Post-Processing'. See as7056_57_config_pp_t
AS7056_57_REG_GROUP_ID_FIFO	Group ID of the register group 'FIFO'. See as7056_57_config_fifo_t
AS7056_57_REG_GROUP_ID_NUM	Maximum number of supported register groups

4.4.4.2 as7056_57_sub_sample_flags enum [as7056_57_sub_sample_flags](#)

Channel flags

Enumerator

AS7056_57_SUB_SAMPLE_FLAG_NONE	No sub-sample is selected
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB1	Flag for sub-sample 1 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB2	Flag for sub-sample 2 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB3	Flag for sub-sample 3 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB4	Flag for sub-sample 4 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB5	Flag for sub-sample 5 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB6	Flag for sub-sample 6 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB7	Flag for sub-sample 7 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ1_SUB8	Flag for sub-sample 8 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB1	Flag for sub-sample 1 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB2	Flag for sub-sample 2 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB3	Flag for sub-sample 3 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB4	Flag for sub-sample 4 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB5	Flag for sub-sample 5 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB6	Flag for sub-sample 6 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB7	Flag for sub-sample 7 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ1_SUB8	Flag for sub-sample 8 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB1	Flag for sub-sample 1 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB2	Flag for sub-sample 2 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB3	Flag for sub-sample 3 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD1_SEQ2_SUB4	Flag for sub-sample 4 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB1	Flag for sub-sample 1 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB2	Flag for sub-sample 2 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB3	Flag for sub-sample 3 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_MOD2_SEQ2_SUB4	Flag for sub-sample 4 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_FLAG_SAR	Flag for SAR status information
AS7056_57_SUB_SAMPLE_FLAG_AOC	Flag for AOC status information
AS7056_57_SUB_SAMPLE_FLAG_SEQ1	Mask of all sequencer 1 sub-samples
AS7056_57_SUB_SAMPLE_FLAG_SEQ2	Mask of all sequencer 2 sub-samples

4.4.4.3 as7056_57_sub_sample_ids enum [as7056_57_sub_sample_ids](#)

Sub-sample ID

Enumerator

AS7056_57_SUB_SAMPLE_ID_DISABLED	No sub-sample ID selected
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB1	Sub-sample ID for sub-sample 1 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB2	Sub-sample ID for sub-sample 2 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB3	Sub-sample ID for sub-sample 3 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB4	Sub-sample ID for sub-sample 4 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB5	Sub-sample ID for sub-sample 5 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB6	Sub-sample ID for sub-sample 6 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB7	Sub-sample ID for sub-sample 7 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ1_SUB8	Sub-sample ID for sub-sample 8 of modulator 1 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB1	Sub-sample ID for sub-sample 1 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB2	Sub-sample ID for sub-sample 2 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB3	Sub-sample ID for sub-sample 3 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB4	Sub-sample ID for sub-sample 4 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB5	Sub-sample ID for sub-sample 5 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB6	Sub-sample ID for sub-sample 6 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB7	Sub-sample ID for sub-sample 7 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ1_SUB8	Sub-sample ID for sub-sample 8 of modulator 2 and sequence 1
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB1	Sub-sample ID for sub-sample 1 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB2	Sub-sample ID for sub-sample 2 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB3	Sub-sample ID for sub-sample 3 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MOD1_SEQ2_SUB4	Sub-sample ID for sub-sample 4 of modulator 1 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB1	Sub-sample ID for sub-sample 1 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB2	Sub-sample ID for sub-sample 2 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB3	Sub-sample ID for sub-sample 3 of modulator 2 and sequence 2

Enumerator

AS7056_57_SUB_SAMPLE_ID_MOD2_SEQ2_SUB4	Sub-sample ID for sub-sample 4 of modulator 2 and sequence 2
AS7056_57_SUB_SAMPLE_ID_MAX	Maximum supported sub-sample number

4.4.4.4 as7056_57_fifo_data_marker enum as7056_57_fifo_data_marker

First two bits of the measurement data, which describes the type of the sample

Enumerator

AS7056_57_FIFO_DATA_MARKER_MOD1_SEQ1_FIRST	Defines first sub-sample of modulator 1 and sequence 1
AS7056_57_FIFO_DATA_MARKER_MOD1_SEQ2_FIRST	Defines first sub-sample of modulator 1 and sequence 2
AS7056_57_FIFO_DATA_MARKER_MOD1_OTHER	Other sub-samples on modulator 1
AS7056_57_FIFO_DATA_MARKER_MOD2_SEQ1_FIRST	Defines first sub-sample of modulator 2 and sequence 1
AS7056_57_FIFO_DATA_MARKER_MOD2_SEQ2_FIRST	Defines first sub-sample of modulator 2 and sequence 2
AS7056_57_FIFO_DATA_MARKER_MOD2_OTHER	Other sub-samples on modulator 2
AS7056_57_FIFO_DATA_MARKER_SAR	Defines SAR status package with PD-offset information
AS7056_57_FIFO_DATA_MARKER_AOC	Status marker for AOC information

4.4.4.5 as7056_57_seq_ids enum as7056_57_seq_ids

Sequence IDs

Enumerator

AS7056_57_ID_SEQ_NONE	ID for unknown sequence
AS7056_57_ID_SEQ1	ID for sequence 1
AS7056_57_ID_SEQ2	ID for sequence 2

4.4.4.6 as7056_57_reg_addresses enum as7056_57_reg_addresses

Register definition of AS7056/57

Enumerator

AS7056_57_REGADDR_CONTROL	Register CONTROL
AS7056_57_REGADDR_CGB_CFG	Register CGB_CFG
AS7056_57_REGADDR_INT_CFG	Register INT_CFG
AS7056_57_REGADDR_CSXN_CFG	Register CSXN_CFG
AS7056_57_REGADDR_IO_CFG	Register IO_CFG
AS7056_57_REGADDR_REF_CFGA	Register REF_CFGA
AS7056_57_REGADDR_REF_CFGB	Register REF_CFGB
AS7056_57_REGADDR_MOD_CFGA	Register MOD_CFGA
AS7056_57_REGADDR_MOD_CFGB	Register MOD_CFGB
AS7056_57_REGADDR_MOD1_CFGA	Register MOD1_CFGA
AS7056_57_REGADDR_MOD1_CFGB	Register MOD1_CFGB
AS7056_57_REGADDR_MOD1_CFGC	Register MOD1_CFGC
AS7056_57_REGADDR_MOD1_CFGD	Register MOD1_CFGD
AS7056_57_REGADDR_MOD1_CFGE	Register MOD1_CFGE
AS7056_57_REGADDR_MOD1_CFGF	Register MOD1_CFGF
AS7056_57_REGADDR_MOD2_CFGA	Register MOD2_CFGA
AS7056_57_REGADDR_MOD2_CFGB	Register MOD2_CFGB
AS7056_57_REGADDR_MOD2_CFGC	Register MOD2_CFGC
AS7056_57_REGADDR_MOD2_CFGD	Register MOD2_CFGD
AS7056_57_REGADDR_MOD2_CFGE	Register MOD2_CFGE
AS7056_57_REGADDR_MOD2_CFGF	Register MOD2_CFGF
AS7056_57_REGADDR_VCSEL_CFG	Register VCSEL_CFG
AS7056_57_REGADDR_SEQ1_LED1_CURR	Register SEQ1_LED1_CURR
AS7056_57_REGADDR_SEQ2_LED1_CURR	Register SEQ2_LED1_CURR
AS7056_57_REGADDR_SEQ1_LED2_CURR	Register SEQ1_LED2_CURR
AS7056_57_REGADDR_SEQ2_LED2_CURR	Register SEQ2_LED2_CURR
AS7056_57_REGADDR_SEQ1_LED3_CURR	Register SEQ1_LED3_CURR
AS7056_57_REGADDR_SEQ2_LED3_CURR	Register SEQ2_LED3_CURR
AS7056_57_REGADDR_LED_SEQ1_SUB12	Register LED_SEQ1_SUB12
AS7056_57_REGADDR_LED_SEQ1_SUB34	Register LED_SEQ1_SUB34
AS7056_57_REGADDR_LED_SEQ1_SUB56	Register LED_SEQ1_SUB56
AS7056_57_REGADDR_LED_SEQ1_SUB78	Register LED_SEQ1_SUB78
AS7056_57_REGADDR_LED_SEQ2_SUB12	Register LED_SEQ2_SUB12
AS7056_57_REGADDR_LED_SEQ2_SUB34	Register LED_SEQ2_SUB34
AS7056_57_REGADDR_LED_LOWVDS_WAIT	Register LED_LOWVDS_WAIT
AS7056_57_REGADDR_PP_CFG	Register PP_CFG
AS7056_57_REGADDR_SEQ1_SUB12_PP	Register SEQ1_SUB12_PP
AS7056_57_REGADDR_SEQ1_SUB34_PP	Register SEQ1_SUB34_PP
AS7056_57_REGADDR_SEQ1_SUB56_PP	Register SEQ1_SUB56_PP
AS7056_57_REGADDR_SEQ1_SUB78_PP	Register SEQ1_SUB78_PP
AS7056_57_REGADDR_SEQ2_SUB12_PP	Register SEQ2_SUB12_PP
AS7056_57_REGADDR_SEQ2_SUB34_PP	Register SEQ2_SUB34_PP
AS7056_57_REGADDR_IRQ_ENABLE	Register IRQ_ENABLE

Enumerator

AS7056_57_REGADDR_SEQ_SAMPLE	Register SEQ_SAMPLE
AS7056_57_REGADDR_SEQ_SUB_WAIT	Register SEQ_SUB_WAIT
AS7056_57_REGADDR_SEQ_MODCONF	Register SEQ_MODCONF
AS7056_57_REGADDR_SEQ_CONFIG	Register SEQ_CONFIG
AS7056_57_REGADDR_SEQ_SAR_WAIT	Register SEQ_SAR_WAIT
AS7056_57_REGADDR_SEQ_LED_INIT	Register SEQ_LED_INIT
AS7056_57_REGADDR_SEQ_FREQ_L	Register SEQ_FREQ_L
AS7056_57_REGADDR_SEQ_FREQ_H	Register SEQ_FREQ_H
AS7056_57_REGADDR_SEQ1_FREQDIV_L	Register SEQ1_FREQDIV_L
AS7056_57_REGADDR_SEQ1_FREQDIV_H	Register SEQ1_FREQDIV_H
AS7056_57_REGADDR_SEQ2_FREQDIV_L	Register SEQ2_FREQDIV_L
AS7056_57_REGADDR_SEQ2_FREQDIV_H	Register SEQ2_FREQDIV_H
AS7056_57_REGADDR_MOD1_SEQ1_SUB_EN	Register MOD1_SEQ1_SUB_EN
AS7056_57_REGADDR_MOD1_SEQ2_SUB_EN	Register MOD1_SEQ2_SUB_EN
AS7056_57_REGADDR_MOD2_SEQ1_SUB_EN	Register MOD2_SEQ1_SUB_EN
AS7056_57_REGADDR_MOD2_SEQ2_SUB_EN	Register MOD2_SEQ2_SUB_EN
AS7056_57_REGADDR_SEQ1_MODE_A	Register SEQ1_MODE_A
AS7056_57_REGADDR_SEQ1_MODE_B	Register SEQ1_MODE_B
AS7056_57_REGADDR_SEQ2_MODE	Register SEQ2_MODE
AS7056_57_REGADDR_PD_SEQ1_SUB1	Register PD_SEQ1_SUB1
AS7056_57_REGADDR_PD_SEQ1_SUB2	Register PD_SEQ1_SUB2
AS7056_57_REGADDR_PD_SEQ1_SUB3	Register PD_SEQ1_SUB3
AS7056_57_REGADDR_PD_SEQ1_SUB4	Register PD_SEQ1_SUB4
AS7056_57_REGADDR_PD_SEQ1_SUB5	Register PD_SEQ1_SUB5
AS7056_57_REGADDR_PD_SEQ1_SUB6	Register PD_SEQ1_SUB6
AS7056_57_REGADDR_PD_SEQ1_SUB7	Register PD_SEQ1_SUB7
AS7056_57_REGADDR_PD_SEQ1_SUB8	Register PD_SEQ1_SUB8
AS7056_57_REGADDR_PD_SEQ2_SUB1	Register PD_SEQ2_SUB1
AS7056_57_REGADDR_PD_SEQ2_SUB2	Register PD_SEQ2_SUB2
AS7056_57_REGADDR_PD_SEQ2_SUB3	Register PD_SEQ2_SUB3
AS7056_57_REGADDR_PD_SEQ2_SUB4	Register PD_SEQ2_SUB4
AS7056_57_REGADDR_PDSEL_CFG	Register PDSEL_CFG
AS7056_57_REGADDR_SEQ1_SINC_CFGA	Register SEQ1_SINC_CFGA
AS7056_57_REGADDR_SEQ1_SINC_CFGB	Register SEQ1_SINC_CFGB
AS7056_57_REGADDR_SEQ1_SINC_CFGC	Register SEQ1_SINC_CFGC
AS7056_57_REGADDR_SEQ2_SINC_CFGA	Register SEQ2_SINC_CFGA
AS7056_57_REGADDR_SEQ2_SINC_CFGB	Register SEQ2_SINC_CFGB
AS7056_57_REGADDR_SEQ2_SINC_CFGC	Register SEQ2_SINC_CFGC
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB1	Register AOC_MOD1_SEQ1_SUB1
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB2	Register AOC_MOD1_SEQ1_SUB2
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB3	Register AOC_MOD1_SEQ1_SUB3
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB4	Register AOC_MOD1_SEQ1_SUB4
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB5	Register AOC_MOD1_SEQ1_SUB5

Enumerator

AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB6	Register AOC_MOD1_SEQ1_SUB6
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB7	Register AOC_MOD1_SEQ1_SUB7
AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB8	Register AOC_MOD1_SEQ1_SUB8
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB1	Register AOC_MOD1_SEQ2_SUB1
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB2	Register AOC_MOD1_SEQ2_SUB2
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB3	Register AOC_MOD1_SEQ2_SUB3
AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB4	Register AOC_MOD1_SEQ2_SUB4
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB1	Register AOC_MOD2_SEQ1_SUB1
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB2	Register AOC_MOD2_SEQ1_SUB2
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB3	Register AOC_MOD2_SEQ1_SUB3
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB4	Register AOC_MOD2_SEQ1_SUB4
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB5	Register AOC_MOD2_SEQ1_SUB5
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB6	Register AOC_MOD2_SEQ1_SUB6
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB7	Register AOC_MOD2_SEQ1_SUB7
AS7056_57_REGADDR_AOC_MOD2_SEQ1_SUB8	Register AOC_MOD2_SEQ1_SUB8
AS7056_57_REGADDR_AOC_LED OFF	Register AOC_LED OFF
AS7056_57_REGADDR_AOC_CFG	Register AOC_CFG
AS7056_57_REGADDR_AOC_MOD1_THH	Register AOC_MOD1_THH
AS7056_57_REGADDR_AOC_MOD1_THL	Register AOC_MOD1_THL
AS7056_57_REGADDR_AOC_MOD2_THH	Register AOC_MOD2_THH
AS7056_57_REGADDR_AOC_MOD2_THL	Register AOC_MOD2_THL
AS7056_57_REGADDR_AOC_SAR_THRES	Register AOC_SAR_THRES
AS7056_57_REGADDR_MOD1_SEQ1_AOC_EN	Register MOD1_SEQ1_AOC_EN
AS7056_57_REGADDR_MOD2_SEQ1_AOC_EN	Register MOD2_SEQ1_AOC_EN
AS7056_57_REGADDR_PROX_CFG	Register PROX_CFG
AS7056_57_REGADDR_PROX_OVS	Register PROX_OVS
AS7056_57_REGADDR_PROX_THH_L	Register PROX_THH_L
AS7056_57_REGADDR_PROX_THH_H	Register PROX_THH_H
AS7056_57_REGADDR_PROX_THL_L	Register PROX_THL_L
AS7056_57_REGADDR_PROX_THL_H	Register PROX_THL_H
AS7056_57_REGADDR_STANDBY_ON	Register STANDBY_ON
AS7056_57_REGADDR_STANDBY_EN1	Register STANDBY_EN1
AS7056_57_REGADDR_STANDBY_EN2	Register STANDBY_EN2
AS7056_57_REGADDR_STANDBY_EN3	Register STANDBY_EN3
AS7056_57_REGADDR_STANDBY_EN4	Register STANDBY_EN4
AS7056_57_REGADDR_STANDBY_EN5	Register STANDBY_EN5
AS7056_57_REGADDR_STANDBY_EN6	Register STANDBY_EN6
AS7056_57_REGADDR_STANDBY_EN7	Register STANDBY_EN7
AS7056_57_REGADDR_FIFO_THRESHOLD	Register FIFO_THRESHOLD
AS7056_57_REGADDR_FIFO_CTRL	Register FIFO_CTRL
AS7056_57_REGADDR_PRODUCT_ID	Register PRODUCT_ID
AS7056_57_REGADDR_SILICON_ID	Register SILICON_ID
AS7056_57_REGADDR_REVISION	Register REVISION

Enumerator

AS7056_57_REGADDR_CHIP_CTRL	Register CHIP_CTRL
AS7056_57_REGADDR_SEQ_START	Register SEQ_START
AS7056_57_REGADDR_STATUS_CGBA	Register STATUS_CGBA
AS7056_57_REGADDR_STATUS_CGBB	Register STATUS_CGBB
AS7056_57_REGADDR_STATUS_SEQ	Register STATUS_SEQ
AS7056_57_REGADDR_STATUS_LED	Register STATUS_LED
AS7056_57_REGADDR_STATUS_ASAT	Register STATUS_ASAT
AS7056_57_REGADDR_STATUS_VCSEL	Register STATUS_VCSEL
AS7056_57_REGADDR_STATUS_PROX	Register STATUS_PROX
AS7056_57_REGADDR_STATUS	Register STATUS
AS7056_57_REGADDR_FIFO_LEVEL0	Register FIFO_LEVEL0
AS7056_57_REGADDR_FIFO_LEVEL1	Register FIFO_LEVEL1
AS7056_57_REGADDR_FIFOL	Register FIFOL
AS7056_57_REGADDR_FIFOM	Register FIFOM
AS7056_57_REGADDR_FIFOH	Register FIFOH

4.4.4.7 as7056_57_chip_events enum [as7056_57_chip_events](#)

Enumeration of chip event flags

Enumerator

AS7056_57_CHIP_EVENT_PROX_LOW_OFF	Mask for bit 'prox_low_off' in register 'STATUS_PROX'
AS7056_57_CHIP_EVENT_PROX_LOW_ON	Mask for bit 'prox_low_on' in register 'STATUS_PROX'
AS7056_57_CHIP_EVENT_PROX_LOW	Mask for bit 'prox_low' in register 'STATUS_PROX'
AS7056_57_CHIP_EVENT_PROX_HIGH_OFF	Mask for bit 'prox_high_off' in register 'STATUS_PROX'
AS7056_57_CHIP_EVENT_PROX_HIGH_ON	Mask for bit 'prox_high_on' in register 'STATUS_PROX'
AS7056_57_CHIP_EVENT_PROX_HIGH	Mask for bit 'prox_high' in register 'STATUS_PROX'
AS7056_57_CHIP_EVENT_VCSEL_WD	Mask for bit 'vcsel_wd' in register 'STATUS_VCSEL'
AS7056_57_CHIP_EVENT_VCSEL_SHORT_VDD	Mask for bit 'vcsel_short_vdd' in register 'STATUS_VCSEL'
AS7056_57_CHIP_EVENT_VCSEL_SHORT_VSS	Mask for bit 'vcsel_short_vss' in register 'STATUS_VCSEL'
AS7056_57_CHIP_EVENT_MOD2_ASAT	Mask for bits 'mod2_asat' in register 'STATUS_ASAT'
AS7056_57_CHIP_EVENT_MOD1_ASAT	Mask for bits 'mod1_asat' in register 'STATUS_ASAT'
AS7056_57_CHIP_EVENT_LED_LOWVDS	Mask for bits 'led_lowvds' in register 'STATUS_LED'
AS7056_57_CHIP_EVENT_SEQ_ERROR	Mask for bit 'seq_error' in register 'STATUS_SEQ'
AS7056_57_CHIP_EVENT_SEQ_END	Mask for bit 'seq_end' in register 'STATUS_SEQ'
AS7056_57_CHIP_EVENT_PROX_BITS_SHIFT	Shift of proximity status bits
AS7056_57_CHIP_EVENT_VCSEL_BITS_SHIFT	Shift of VCSEL status bits
AS7056_57_CHIP_EVENT_ASAT_BITS_SHIFT	Shift of analog saturation status bits
AS7056_57_CHIP_EVENT_LED_BITS_SHIFT	Shift of LED status bits
AS7056_57_CHIP_EVENT_SEQ_BITS_SHIFT	Shift of sequence status bits

5 Data Structure Documentation

5.1 as7056_57_config_aoc_t::aoc_regs Struct Reference

Data Fields

- uint8_t [aoc_mod1_seq1_sub1](#)
- uint8_t [aoc_mod1_seq1_sub2](#)
- uint8_t [aoc_mod1_seq1_sub3](#)
- uint8_t [aoc_mod1_seq1_sub4](#)
- uint8_t [aoc_mod1_seq1_sub5](#)
- uint8_t [aoc_mod1_seq1_sub6](#)
- uint8_t [aoc_mod1_seq1_sub7](#)
- uint8_t [aoc_mod1_seq1_sub8](#)
- uint8_t [aoc_mod1_seq2_sub1](#)
- uint8_t [aoc_mod1_seq2_sub2](#)
- uint8_t [aoc_mod1_seq2_sub3](#)
- uint8_t [aoc_mod1_seq2_sub4](#)
- uint8_t [aoc_mod2_seq1_sub1](#)
- uint8_t [aoc_mod2_seq1_sub2](#)
- uint8_t [aoc_mod2_seq1_sub3](#)
- uint8_t [aoc_mod2_seq1_sub4](#)
- uint8_t [aoc_mod2_seq1_sub5](#)
- uint8_t [aoc_mod2_seq1_sub6](#)
- uint8_t [aoc_mod2_seq1_sub7](#)
- uint8_t [aoc_mod2_seq1_sub8](#)
- uint8_t [aoc_ledoff](#)
- uint8_t [aoc_cfg](#)
- uint8_t [aoc_mod1_thh](#)
- uint8_t [aoc_mod1_thl](#)
- uint8_t [aoc_mod2_thh](#)
- uint8_t [aoc_mod2_thl](#)
- uint8_t [aoc_sar_thres](#)
- uint8_t [mod1_seq1_aoc_en](#)
- uint8_t [mod2_seq1_aoc_en](#)

5.1.1 Detailed Description

Register content for AOC configuration

5.1.2 Field Documentation

5.1.2.1 aoc_mod1_seq1_sub1 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub1

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB1](#)

5.1.2.2 aoc_mod1_seq1_sub2 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub2

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB2](#)

5.1.2.3 aoc_mod1_seq1_sub3 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub3

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB3](#)

5.1.2.4 aoc_mod1_seq1_sub4 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub4

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB4](#)

5.1.2.5 aoc_mod1_seq1_sub5 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub5

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB5](#)

5.1.2.6 aoc_mod1_seq1_sub6 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub6

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB6](#)

5.1.2.7 aoc_mod1_seq1_sub7 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub7

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB7](#)

5.1.2.8 aoc_mod1_seq1_sub8 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq1_sub8

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB8](#)

5.1.2.9 aoc_mod1_seq2_sub1 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq2_sub1

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB1](#)

5.1.2.10 aoc_mod1_seq2_sub2 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq2_sub2

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB2](#)

5.1.2.11 aoc_mod1_seq2_sub3 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq2_sub3

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB3](#)

5.1.2.12 aoc_mod1_seq2_sub4 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_seq2_sub4

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ2_SUB4](#)

5.1.2.13 aoc_mod2_seq1_sub1 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub1

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB1](#)

5.1.2.14 aoc_mod2_seq1_sub2 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub2

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB2](#)

5.1.2.15 aoc_mod2_seq1_sub3 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub3

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB3](#)

5.1.2.16 aoc_mod2_seq1_sub4 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub4

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB4](#)

5.1.2.17 aoc_mod2_seq1_sub5 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub5

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB5](#)

5.1.2.18 aoc_mod2_seq1_sub6 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub6

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB6](#)

5.1.2.19 aoc_mod2_seq1_sub7 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub7

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB7](#)

5.1.2.20 aoc_mod2_seq1_sub8 uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_seq1_sub8

Content of register [AS7056_57_REGADDR_AOC_MOD1_SEQ1_SUB8](#)

5.1.2.21 aoc_ledoff uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_ledoff

Content of register [AS7056_57_REGADDR_AOC_LED OFF](#)

5.1.2.22 aoc_cfg uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_cfg

Content of register [AS7056_57_REGADDR_AOC_CFG](#)

5.1.2.23 aoc_mod1_thh uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_thh

Content of register [AS7056_57_REGADDR_AOC_MOD1_THH](#)

5.1.2.24 aoc_mod1_thl uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod1_thl

Content of register [AS7056_57_REGADDR_AOC_MOD1_THL](#)

5.1.2.25 aoc_mod2_thh uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_thh

Content of register [AS7056_57_REGADDR_AOC_MOD2_THH](#)

5.1.2.26 aoc_mod2_thl uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_mod2_thl

Content of register [AS7056_57_REGADDR_AOC_MOD2_THL](#)

5.1.2.27 aoc_sar_thres uint8_t as7056_57_config_aoc_t::aoc_regs::aoc_sar_thres

Content of register [AS7056_57_REGADDR_AOC_SAR_THRES](#)

5.1.2.28 mod1_seq1_aoc_en uint8_t as7056_57_config_aoc_t::aoc_regs::mod1_seq1_aoc_en

Content of register [AS7056_57_REGADDR_MOD1_SEQ1_AOC_EN](#)

5.1.2.29 mod2_seq1_aoc_en uint8_t as7056_57_config_aoc_t::aoc_regs::mod2_seq1_aoc_en

Content of register [AS7056_57_REGADDR_MOD2_SEQ1_AOC_EN](#)

5.2 as7056_57_config_aoc_t Union Reference

Data Structures

- struct [aoc_regs](#)

Data Fields

- struct [as7056_57_config_aoc_t::aoc_regs](#) reg_vals
- uint8_t [reg_buffer](#) [sizeof(struct [aoc_regs](#))]

5.2.1 Detailed Description

Register group for configuration of Automatic Offset Control (AOC)

5.2.2 Field Documentation

5.2.2.1 reg_vals struct `as7056_57_config_aoc_t::aoc_regs` `as7056_57_config_aoc_t::reg_vals`

Register content for AOC configuration

5.2.2.2 reg_buffer `uint8_t` `as7056_57_config_aoc_t::reg_buffer[sizeof(struct aoc_regs)]`

Register content for AOC configuration

5.3 as7056_57_config_ctrl_t Union Reference

Data Structures

- struct `ctrl_regs`

Data Fields

- struct `as7056_57_config_ctrl_t::ctrl_regs` `reg_vals`
- `uint8_t` `reg_buffer` `[sizeof(struct ctrl_regs)]`

5.3.1 Detailed Description

Register group for configuration of GPIOs and clock

5.3.2 Field Documentation

5.3.2.1 reg_vals struct `as7056_57_config_ctrl_t::ctrl_regs` `as7056_57_config_ctrl_t::reg_vals`

Register content for GPIO configuration

5.3.2.2 reg_buffer `uint8_t` `as7056_57_config_ctrl_t::reg_buffer[sizeof(struct ctrl_regs)]`

Register content for CTRL configuration

5.4 as7056_57_config_fifo_t Union Reference

Data Structures

- struct `fifo_regs`

Data Fields

- struct [as7056_57_config_fifo_t::fifo_regs](#) [reg_vals](#)
- [uint8_t](#) [reg_buffer](#) [sizeof(struct [fifo_regs](#))]

5.4.1 Detailed Description

Register group for configuration of the FIFO handling

5.4.2 Field Documentation

5.4.2.1 [reg_vals](#) struct [as7056_57_config_fifo_t::fifo_regs](#) [as7056_57_config_fifo_t::reg_vals](#)

Register content for FIFO configuration

5.4.2.2 [reg_buffer](#) [uint8_t](#) [as7056_57_config_fifo_t::reg_buffer](#)[sizeof(struct [fifo_regs](#))]

Register content for FIFO configuration

5.5 [as7056_57_config_led_t](#) Union Reference

Data Structures

- struct [led_regs](#)

Data Fields

- struct [as7056_57_config_led_t::led_regs](#) [reg_vals](#)
- [uint8_t](#) [reg_buffer](#) [sizeof(struct [led_regs](#))]

5.5.1 Detailed Description

Register group for LED configuration

5.5.2 Field Documentation

5.5.2.1 reg_vals struct `as7056_57_config_led_t::led_regs` `as7056_57_config_led_t::reg_vals`

Register content for LED configuration

5.5.2.2 reg_buffer `uint8_t` `as7056_57_config_led_t::reg_buffer[sizeof(struct led_regs)]`

Register content for LED configuration

5.6 as7056_57_config_mod_t Union Reference

Data Structures

- struct `mod_regs`

Data Fields

- struct `as7056_57_config_mod_t::mod_regs` `reg_vals`
- `uint8_t` `reg_buffer` `[sizeof(struct mod_regs)]`

5.6.1 Detailed Description

Register group for configuration of the modulator

5.6.2 Field Documentation

5.6.2.1 reg_vals struct `as7056_57_config_mod_t::mod_regs` `as7056_57_config_mod_t::reg_vals`

Register content for MOD configuration

5.6.2.2 reg_buffer `uint8_t` `as7056_57_config_mod_t::reg_buffer[sizeof(struct mod_regs)]`

Register content for MOD configuration

5.7 as7056_57_config_pd_t Union Reference

Data Structures

- struct `pd_regs`

Data Fields

- struct [as7056_57_config_pd_t::pd_regs](#) [reg_vals](#)
- uint8_t [reg_buffer](#) [sizeof(struct [pd_regs](#))]

5.7.1 Detailed Description

Register group for configuration of the photodiodes

5.7.2 Field Documentation

5.7.2.1 [reg_vals](#) struct [as7056_57_config_pd_t::pd_regs](#) [as7056_57_config_pd_t::reg_vals](#)

Register content for photodiodes configuration

5.7.2.2 [reg_buffer](#) uint8_t [as7056_57_config_pd_t::reg_buffer](#)[sizeof(struct [pd_regs](#))]

Register content for photodiodes configuration

5.8 [as7056_57_config_pp_t](#) Union Reference

Data Structures

- struct [pp_regs](#)

Data Fields

- struct [as7056_57_config_pp_t::pp_regs](#) [reg_vals](#)
- uint8_t [reg_buffer](#) [sizeof(struct [pp_regs](#))]

5.8.1 Detailed Description

Register group for configuration of post-processing engine

5.8.2 Field Documentation

5.8.2.1 reg_vals struct [as7056_57_config_pp_t::pp_regs](#) as7056_57_config_pp_t::reg_vals

Register content for post-processing configuration

5.8.2.2 reg_buffer uint8_t as7056_57_config_pp_t::reg_buffer[sizeof(struct [pp_regs](#))]

Register content for post-processing configuration

5.9 as7056_57_config_prox_t Union Reference

Data Structures

- struct [prox_regs](#)

Data Fields

- struct [as7056_57_config_prox_t::prox_regs](#) reg_vals
- uint8_t [reg_buffer](#) [sizeof(struct [prox_regs](#))]

5.9.1 Detailed Description

Register group for configuration the proximity component

5.9.2 Field Documentation

5.9.2.1 reg_vals struct [as7056_57_config_prox_t::prox_regs](#) as7056_57_config_prox_t::reg_vals

Register content for PROX configuration

5.9.2.2 reg_buffer uint8_t as7056_57_config_prox_t::reg_buffer[sizeof(struct [prox_regs](#))]

Register content for PROX configuration

5.10 as7056_57_config_ref_t Union Reference

Data Structures

- struct [ref_regs](#)

Data Fields

- struct [as7056_57_config_ref_t::ref_regs](#) [reg_vals](#)
- uint8_t [reg_buffer](#) [sizeof(struct [ref_regs](#))]

5.10.1 Detailed Description

Register group for configuration of the reference registers

5.10.2 Field Documentation

5.10.2.1 [reg_vals](#) struct [as7056_57_config_ref_t::ref_regs](#) [as7056_57_config_ref_t::reg_vals](#)

Register content for REF-register configuration

5.10.2.2 [reg_buffer](#) uint8_t [as7056_57_config_ref_t::reg_buffer](#)[sizeof(struct [ref_regs](#))]

Register content for REF-register configuration

5.11 [as7056_57_config_seq_t](#) Union Reference

Data Structures

- struct [seq_regs](#)

Data Fields

- struct [as7056_57_config_seq_t::seq_regs](#) [reg_vals](#)
- uint8_t [reg_buffer](#) [sizeof(struct [seq_regs](#))]

5.11.1 Detailed Description

Register group for configuration of the sequencer

5.11.2 Field Documentation

5.11.2.1 reg_vals struct [as7056_57_config_seq_t::seq_regs](#) as7056_57_config_seq_t::reg_vals

Register content for sequencer configuration

5.11.2.2 reg_buffer uint8_t as7056_57_config_seq_t::reg_buffer[sizeof(struct [seq_regs](#))]

Register content for sequencer configuration

5.12 as7056_57_config_sinc_t Union Reference

Data Structures

- struct [sinc_regs](#)

Data Fields

- struct [as7056_57_config_sinc_t::sinc_regs](#) reg_vals
- uint8_t [reg_buffer](#) [sizeof(struct [sinc_regs](#))]

5.12.1 Detailed Description

Register group for configuration of the SINC filter

5.12.2 Field Documentation

5.12.2.1 reg_vals struct [as7056_57_config_sinc_t::sinc_regs](#) as7056_57_config_sinc_t::reg_vals

Register content for SINC filter configuration

5.12.2.2 reg_buffer uint8_t as7056_57_config_sinc_t::reg_buffer[sizeof(struct [sinc_regs](#))]

Register content for SINC filter configuration

5.13 as7056_57_config_standby_t Union Reference

Data Structures

- struct [standby_regs](#)

Data Fields

- struct [as7056_57_config_standby_t::standby_regs](#) [reg_vals](#)
- uint8_t [reg_buffer](#) [sizeof(struct [standby_regs](#))]

5.13.1 Detailed Description

Register group for configuration for STANDBY mode

5.13.2 Field Documentation

5.13.2.1 [reg_vals](#) struct [as7056_57_config_standby_t::standby_regs](#) [as7056_57_config_standby_t](#)↔
::[reg_vals](#)

Register content for STANDBY configuration

5.13.2.2 [reg_buffer](#) uint8_t [as7056_57_config_standby_t::reg_buffer](#)[sizeof(struct [standby_regs](#))]

Register content for STANDBY configuration

5.14 [as7056_57_extract_metadata_t](#) Struct Reference

Data Fields

- uint32_t [fifo_map](#)
- struct [extract_metadata](#) [old_metadata](#)
- struct [extract_metadata](#) [new_metadata](#)
- uint8_t [copy_new_to_old](#)

5.14.1 Detailed Description

Metadata for extraction function

5.14.2 Field Documentation

5.14.2.1 [fifo_map](#) uint32_t [as7056_57_extract_metadata_t::fifo_map](#)

Currently available sub-samples inside the FIFO data stream. See [as7056_57_sub_sample_flags](#) and [as7056_57_meas_config_t](#).

5.14.2.2 old_metadata `struct extract_metadata as7056_57_extract_metadata_t::old_metadata`

Metadata of old data.

5.14.2.3 new_metadata `struct extract_metadata as7056_57_extract_metadata_t::new_metadata`

Metadata of new data.

5.14.2.4 copy_new_to_old `uint8_t as7056_57_extract_metadata_t::copy_new_to_old`

Non-zero value: old meta data will be overwritten by new metadata.

5.15 as7056_57_interrupt_enable_t Struct Reference

Data Fields

- `uint8_t sequencer`: 1
- `uint8_t fifo_threshold`: 1
- `uint8_t fifo_overflow`: 1
- `uint8_t led_lowvds`: 1
- `uint8_t asat`: 1
- `uint8_t vcsl`: 1
- `uint8_t prox_off`: 1
- `uint8_t prox_on`: 1

5.15.1 Detailed Description

IRQ enable register bits

5.15.2 Field Documentation

5.15.2.1 sequencer `uint8_t as7056_57_interrupt_enable_t::sequencer`

Enable interrupt for sequencer: Error or stop

5.15.2.2 fifo_threshold `uint8_t as7056_57_interrupt_enable_t::fifo_threshold`

Enable interrupt if FIFO level is greater than FIFO threshold

5.15.2.3 **fifo_overflow** `uint8_t as7056_57_interrupt_enable_t::fifo_overflow`

Enable interrupt if FIFO overflow is detected

5.15.2.4 **led_lowvds** `uint8_t as7056_57_interrupt_enable_t::led_lowvds`

Enable interrupt if LED current does not reach the expected value

5.15.2.5 **asat** `uint8_t as7056_57_interrupt_enable_t::asat`

Enable interrupt on analog saturation

5.15.2.6 **vcsel** `uint8_t as7056_57_interrupt_enable_t::vcsel`

Enable interrupt on VCSEL error

5.15.2.7 **prox_off** `uint8_t as7056_57_interrupt_enable_t::prox_off`

Enable interrupt on falling edge of PROX_HIGH and PROX_LOW

5.15.2.8 **prox_on** `uint8_t as7056_57_interrupt_enable_t::prox_on`

Enable interrupt on rising edge of PROX_HIGH and PROX_LOW

5.16 **as7056_57_meas_config_t** Struct Reference

Data Fields

- `uint32_t seq1_sample_period_us`
- `uint32_t seq2_sample_period_us`
- `uint32_t fifo_map`
- `uint16_t fifo_threshold`
- `as7056_57_interrupt_enable_t enabled_irqs`
- `uint8_t reserved`
- `as7056_57_sub_sample_ids_t agc_channels` [AGC_MAX_CHANNEL_CNT]

5.16.1 Detailed Description

Measurement settings which can be read out after finished register configuration

5.16.2 Field Documentation

5.16.2.1 seq1_sample_period_us uint32_t as7056_57_meas_config_t::seq1_sample_period_us

Sample period for sequence 1 in microseconds

5.16.2.2 seq2_sample_period_us uint32_t as7056_57_meas_config_t::seq2_sample_period_us

Sample period for sequence 2 in microseconds

5.16.2.3 fifo_map uint32_t as7056_57_meas_config_t::fifo_map

Definition which sub-samples are mapped inside FIFO. (More than one flag can be set) See [as7056_57_sub_sample_flags](#)

5.16.2.4 fifo_threshold uint16_t as7056_57_meas_config_t::fifo_threshold

FIFO threshold, when data shall be read

5.16.2.5 enabled_irqs [as7056_57_interrupt_enable_t](#) as7056_57_meas_config_t::enabled_irqs

Enabled IRQ interrupts

5.16.2.6 reserved uint8_t as7056_57_meas_config_t::reserved

padding bytes

5.16.2.7 agc_channels [as7056_57_sub_sample_ids_t](#) as7056_57_meas_config_t::agc_channels[AGC_M↔AX_CHANNEL_CNT]

Contains the ID of the controlled sub-sample for each AGC channel. If an AGC channel is disabled, the corresponding array item is set to [AS7056_57_SUB_SAMPLE_ID_DISABLED](#).

5.17 as7056_57_version Struct Reference

Data Fields

- uint8_t [major](#)
- uint8_t [minor](#)
- uint8_t [patch](#)

5.17.1 Detailed Description

Version information of the library

5.17.2 Field Documentation

5.17.2.1 major `uint8_t as7056_57_version::major`

Major version

5.17.2.2 minor `uint8_t as7056_57_version::minor`

Minor version

5.17.2.3 patch `uint8_t as7056_57_version::patch`

Patch version

5.18 as7056_57_config_ctrl_t::ctrl_regs Struct Reference

Data Fields

- `uint8_t control`
- `uint8_t cgb_cfg`
- `uint8_t int_cfg`
- `uint8_t csxn_cfg`
- `uint8_t io_cfg`

5.18.1 Detailed Description

Register content for CTRL configuration

5.18.2 Field Documentation

5.18.2.1 control `uint8_t as7056_57_config_ctrl_t::ctrl_regs::control`

Content of register [AS7056_57_REGADDR_CONTROL](#)

5.18.2.2 cgb_cfg `uint8_t as7056_57_config_ctrl_t::ctrl_regs::cgb_cfg`

Content of register [AS7056_57_REGADDR_CGB_CFG](#)

5.18.2.3 int_cfg `uint8_t as7056_57_config_ctrl_t::ctrl_regs::int_cfg`

Content of register [AS7056_57_REGADDR_INT_CFG](#)

5.18.2.4 csxn_cfg `uint8_t as7056_57_config_ctrl_t::ctrl_regs::csxn_cfg`

Content of register [AS7056_57_REGADDR_CSXN_CFG](#)

5.18.2.5 io_cfg `uint8_t as7056_57_config_ctrl_t::ctrl_regs::io_cfg`

Content of register [AS7056_57_REGADDR_IO_CFG](#)

5.19 extract_metadata Struct Reference

Data Fields

- [as7056_57_sub_sample_flags_t mod1_sub](#)
- [as7056_57_sub_sample_flags_t mod2_sub](#)
- [as7056_57_seq_ids_t seq_id](#)

5.19.1 Detailed Description

Metadata description

5.19.2 Field Documentation

5.19.2.1 mod1_sub `as7056_57_sub_sample_flags_t extract_metadata::mod1_sub`

Sub-sample of modulator 1

5.19.2.2 mod2_sub `as7056_57_sub_sample_flags_t extract_metadata::mod2_sub`

Sub-sample of modulator 2

5.19.2.3 seq_id `as7056_57_seq_ids_t extract_metadata::seq_id`

Sequence ID

5.20 as7056_57_config_fifo_t::fifo_regs Struct Reference

Data Fields

- uint8_t [fifo_threshold](#)
- uint8_t [fifo_ctrl](#)

5.20.1 Detailed Description

Register content for FIFO configuration

5.20.2 Field Documentation

5.20.2.1 fifo_threshold uint8_t as7056_57_config_fifo_t::fifo_regs::fifo_threshold

Content of register [AS7056_57_REGADDR_FIFO_THRESHOLD](#)

5.20.2.2 fifo_ctrl uint8_t as7056_57_config_fifo_t::fifo_regs::fifo_ctrl

Content of register [AS7056_57_REGADDR_FIFO_CTRL](#)

5.21 as7056_57_config_led_t::led_regs Struct Reference

Data Fields

- uint8_t [vcsel_cfg](#)
- uint8_t [seq1_led1_curr](#)
- uint8_t [seq2_led1_curr](#)
- uint8_t [seq1_led2_curr](#)
- uint8_t [seq2_led2_curr](#)
- uint8_t [seq1_led3_curr](#)
- uint8_t [seq2_led3_curr](#)
- uint8_t [led_seq1_sub12](#)
- uint8_t [led_seq1_sub34](#)
- uint8_t [led_seq1_sub56](#)
- uint8_t [led_seq1_sub78](#)
- uint8_t [led_seq2_sub12](#)
- uint8_t [led_seq2_sub34](#)
- uint8_t [lowvds_wait](#)

5.21.1 Detailed Description

Register content for LED configuration

5.21.2 Field Documentation

5.21.2.1 vcsel_cfg `uint8_t as7056_57_config_led_t::led_regs::vcsel_cfg`

Content of register [AS7056_57_REGADDR_VCSEL_CFG](#)

5.21.2.2 seq1_led1_curr `uint8_t as7056_57_config_led_t::led_regs::seq1_led1_curr`

Content of register [AS7056_57_REGADDR_SEQ1_LED1_CURR](#)

5.21.2.3 seq2_led1_curr `uint8_t as7056_57_config_led_t::led_regs::seq2_led1_curr`

Content of register [AS7056_57_REGADDR_SEQ2_LED1_CURR](#)

5.21.2.4 seq1_led2_curr `uint8_t as7056_57_config_led_t::led_regs::seq1_led2_curr`

Content of register [AS7056_57_REGADDR_SEQ1_LED2_CURR](#)

5.21.2.5 seq2_led2_curr `uint8_t as7056_57_config_led_t::led_regs::seq2_led2_curr`

Content of register [AS7056_57_REGADDR_SEQ2_LED2_CURR](#)

5.21.2.6 seq1_led3_curr `uint8_t as7056_57_config_led_t::led_regs::seq1_led3_curr`

Content of register [AS7056_57_REGADDR_SEQ1_LED3_CURR](#)

5.21.2.7 seq2_led3_curr `uint8_t as7056_57_config_led_t::led_regs::seq2_led3_curr`

Content of register [AS7056_57_REGADDR_SEQ2_LED3_CURR](#)

5.21.2.8 led_seq1_sub12 `uint8_t as7056_57_config_led_t::led_regs::led_seq1_sub12`

Content of register [AS7056_57_REGADDR_LED_SEQ1_SUB12](#)

5.21.2.9 led_seq1_sub34 uint8_t as7056_57_config_led_t::led_regs::led_seq1_sub34

Content of register [AS7056_57_REGADDR_LED_SEQ1_SUB34](#)

5.21.2.10 led_seq1_sub56 uint8_t as7056_57_config_led_t::led_regs::led_seq1_sub56

Content of register [AS7056_57_REGADDR_LED_SEQ1_SUB56](#)

5.21.2.11 led_seq1_sub78 uint8_t as7056_57_config_led_t::led_regs::led_seq1_sub78

Content of register [AS7056_57_REGADDR_LED_SEQ1_SUB78](#)

5.21.2.12 led_seq2_sub12 uint8_t as7056_57_config_led_t::led_regs::led_seq2_sub12

Content of register [AS7056_57_REGADDR_LED_SEQ2_SUB12](#)

5.21.2.13 led_seq2_sub34 uint8_t as7056_57_config_led_t::led_regs::led_seq2_sub34

Content of register [AS7056_57_REGADDR_LED_SEQ2_SUB34](#)

5.21.2.14 lowvds_wait uint8_t as7056_57_config_led_t::led_regs::lowvds_wait

Content of register [AS7056_57_REGADDR_LED_LOWVDS_WAIT](#)

5.22 as7056_57_config_mod_t::mod_regs Struct Reference

Data Fields

- uint8_t [mod_cfga](#)
- uint8_t [mod_cfgb](#)
- uint8_t [mod1_cfga](#)
- uint8_t [mod1_cfgb](#)
- uint8_t [mod1_cfgc](#)
- uint8_t [mod1_cfgd](#)
- uint8_t [mod1_cfge](#)
- uint8_t [mod1_cfgf](#)
- uint8_t [mod2_cfga](#)
- uint8_t [mod2_cfgb](#)
- uint8_t [mod2_cfgc](#)
- uint8_t [mod2_cfgd](#)
- uint8_t [mod2_cfge](#)
- uint8_t [mod2_cfgf](#)

5.22.1 Detailed Description

Register content for MOD configuration

5.22.2 Field Documentation

5.22.2.1 mod_cfg_a `uint8_t as7056_57_config_mod_t::mod_regs::mod_cfg_a`

Content of register [AS7056_57_REGADDR_MOD_CFGA](#)

5.22.2.2 mod_cfg_b `uint8_t as7056_57_config_mod_t::mod_regs::mod_cfg_b`

Content of register [AS7056_57_REGADDR_MOD_CFGB](#)

5.22.2.3 mod1_cfg_a `uint8_t as7056_57_config_mod_t::mod_regs::mod1_cfg_a`

Content of register [AS7056_57_REGADDR_MOD1_CFGA](#)

5.22.2.4 mod1_cfg_b `uint8_t as7056_57_config_mod_t::mod_regs::mod1_cfg_b`

Content of register [AS7056_57_REGADDR_MOD1_CFGB](#)

5.22.2.5 mod1_cfg_c `uint8_t as7056_57_config_mod_t::mod_regs::mod1_cfg_c`

Content of register [AS7056_57_REGADDR_MOD1_CFGC](#)

5.22.2.6 mod1_cfg_d `uint8_t as7056_57_config_mod_t::mod_regs::mod1_cfg_d`

Content of register [AS7056_57_REGADDR_MOD1_CFGD](#)

5.22.2.7 mod1_cfg_e `uint8_t as7056_57_config_mod_t::mod_regs::mod1_cfg_e`

Content of register [AS7056_57_REGADDR_MOD1_CFGE](#)

5.22.2.8 mod1_cfg_f `uint8_t as7056_57_config_mod_t::mod_regs::mod1_cfg_f`

Content of register [AS7056_57_REGADDR_MOD1_CFGF](#)

5.22.2.9 mod2_cfg_a uint8_t as7056_57_config_mod_t::mod_regs::mod2_cfg_a

Content of register [AS7056_57_REGADDR_MOD2_CFGA](#)

5.22.2.10 mod2_cfg_b uint8_t as7056_57_config_mod_t::mod_regs::mod2_cfg_b

Content of register [AS7056_57_REGADDR_MOD2_CFGB](#)

5.22.2.11 mod2_cfg_c uint8_t as7056_57_config_mod_t::mod_regs::mod2_cfg_c

Content of register [AS7056_57_REGADDR_MOD2_CFGC](#)

5.22.2.12 mod2_cfg_d uint8_t as7056_57_config_mod_t::mod_regs::mod2_cfg_d

Content of register [AS7056_57_REGADDR_MOD2_CFGD](#)

5.22.2.13 mod2_cfg_e uint8_t as7056_57_config_mod_t::mod_regs::mod2_cfg_e

Content of register [AS7056_57_REGADDR_MOD2_CFGE](#)

5.22.2.14 mod2_cfg_f uint8_t as7056_57_config_mod_t::mod_regs::mod2_cfg_f

Content of register [AS7056_57_REGADDR_MOD2_CFGF](#)

5.23 as7056_57_config_pd_t::pd_regs Struct Reference

Data Fields

- uint8_t [pd_seq1_sub1](#)
- uint8_t [pd_seq1_sub2](#)
- uint8_t [pd_seq1_sub3](#)
- uint8_t [pd_seq1_sub4](#)
- uint8_t [pd_seq1_sub5](#)
- uint8_t [pd_seq1_sub6](#)
- uint8_t [pd_seq1_sub7](#)
- uint8_t [pd_seq1_sub8](#)
- uint8_t [pd_seq2_sub1](#)
- uint8_t [pd_seq2_sub2](#)
- uint8_t [pd_seq2_sub3](#)
- uint8_t [pd_seq2_sub4](#)
- uint8_t [pdsel_cfg](#)

5.23.1 Detailed Description

Register content for photodiodes configuration

5.23.2 Field Documentation

5.23.2.1 pd_seq1_sub1 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub1

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB1](#)

5.23.2.2 pd_seq1_sub2 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub2

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB2](#)

5.23.2.3 pd_seq1_sub3 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub3

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB3](#)

5.23.2.4 pd_seq1_sub4 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub4

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB4](#)

5.23.2.5 pd_seq1_sub5 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub5

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB5](#)

5.23.2.6 pd_seq1_sub6 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub6

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB6](#)

5.23.2.7 pd_seq1_sub7 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub7

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB7](#)

5.23.2.8 pd_seq1_sub8 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq1_sub8

Content of register [AS7056_57_REGADDR_PD_SEQ1_SUB8](#)

5.23.2.9 pd_seq2_sub1 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq2_sub1

Content of register [AS7056_57_REGADDR_PD_SEQ2_SUB1](#)

5.23.2.10 pd_seq2_sub2 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq2_sub2

Content of register [AS7056_57_REGADDR_PD_SEQ2_SUB2](#)

5.23.2.11 pd_seq2_sub3 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq2_sub3

Content of register [AS7056_57_REGADDR_PD_SEQ2_SUB3](#)

5.23.2.12 pd_seq2_sub4 uint8_t as7056_57_config_pd_t::pd_regs::pd_seq2_sub4

Content of register [AS7056_57_REGADDR_PD_SEQ2_SUB4](#)

5.23.2.13 pdsel_cfg uint8_t as7056_57_config_pd_t::pd_regs::pdsel_cfg

Content of register [AS7056_57_REGADDR_PDSEL_CFG](#)

5.24 as7056_57_config_pp_t::pp_regs Struct Reference

Data Fields

- uint8_t [pp_cfg](#)
- uint8_t [seq1_sub12_pp](#)
- uint8_t [seq1_sub34_pp](#)
- uint8_t [seq1_sub56_pp](#)
- uint8_t [seq1_sub78_pp](#)
- uint8_t [seq2_sub12_pp](#)
- uint8_t [seq2_sub34_pp](#)

5.24.1 Detailed Description

Register content for post-processing configuration

5.24.2 Field Documentation

5.24.2.1 pp_cfg uint8_t as7056_57_config_pp_t::pp_regs::pp_cfg

Content of register [AS7056_57_REGADDR_PP_CFG](#)

5.24.2.2 seq1_sub12_pp uint8_t as7056_57_config_pp_t::pp_regs::seq1_sub12_pp

Content of register [AS7056_57_REGADDR_SEQ1_SUB12_PP](#)

5.24.2.3 seq1_sub34_pp uint8_t as7056_57_config_pp_t::pp_regs::seq1_sub34_pp

Content of register [AS7056_57_REGADDR_SEQ1_SUB34_PP](#)

5.24.2.4 seq1_sub56_pp uint8_t as7056_57_config_pp_t::pp_regs::seq1_sub56_pp

Content of register [AS7056_57_REGADDR_SEQ1_SUB56_PP](#)

5.24.2.5 seq1_sub78_pp uint8_t as7056_57_config_pp_t::pp_regs::seq1_sub78_pp

Content of register [AS7056_57_REGADDR_SEQ1_SUB78_PP](#)

5.24.2.6 seq2_sub12_pp uint8_t as7056_57_config_pp_t::pp_regs::seq2_sub12_pp

Content of register [AS7056_57_REGADDR_SEQ2_SUB12_PP](#)

5.24.2.7 seq2_sub34_pp uint8_t as7056_57_config_pp_t::pp_regs::seq2_sub34_pp

Content of register [AS7056_57_REGADDR_SEQ2_SUB34_PP](#)

5.25 as7056_57_config_prox_t::prox_regs Struct Reference

Data Fields

- uint8_t [prox_cfg](#)
- uint8_t [prox_ovs](#)
- uint8_t [prox_thh_l](#)
- uint8_t [prox_thh_h](#)
- uint8_t [prox_thl_l](#)
- uint8_t [prox_thl_h](#)

5.25.1 Detailed Description

Register content for proximity

5.25.2 Field Documentation

5.25.2.1 prox_cfg uint8_t as7056_57_config_prox_t::prox_regs::prox_cfg

Content of register [AS7056_57_REGADDR_PROX_CFG](#)

5.25.2.2 prox_ovs uint8_t as7056_57_config_prox_t::prox_regs::prox_ovs

Content of register [AS7056_57_REGADDR_PROX_OVS](#)

5.25.2.3 prox_thh_l uint8_t as7056_57_config_prox_t::prox_regs::prox_thh_l

Content of register [AS7056_57_REGADDR_PROX_THH_L](#)

5.25.2.4 prox_thh_h uint8_t as7056_57_config_prox_t::prox_regs::prox_thh_h

Content of register [AS7056_57_REGADDR_PROX_THH_H](#)

5.25.2.5 prox_thl_l uint8_t as7056_57_config_prox_t::prox_regs::prox_thl_l

Content of register [AS7056_57_REGADDR_PROX_THL_L](#)

5.25.2.6 prox_thl_h uint8_t as7056_57_config_prox_t::prox_regs::prox_thl_h

Content of register [AS7056_57_REGADDR_PROX_THL_H](#)

5.26 as7056_57_config_ref_t::ref_regs Struct Reference

Data Fields

- uint8_t [ref_cfg_a](#)
- uint8_t [ref_cfg_b](#)

5.26.1 Detailed Description

Register content for REF-register configuration

5.26.2 Field Documentation

5.26.2.1 ref_cfg_a uint8_t as7056_57_config_ref_t::ref_regs::ref_cfg_a

Content of register [AS7056_57_REGADDR_REF_CFGA](#)

5.26.2.2 ref_cfg_b uint8_t as7056_57_config_ref_t::ref_regs::ref_cfg_b

Content of register [AS7056_57_REGADDR_REF_CFGB](#)

5.27 as7056_57_config_seq_t::seq_regs Struct Reference

Data Fields

- uint8_t [irq_enable](#)
- uint8_t [seq_sample](#)
- uint8_t [seq_sub_wait](#)
- uint8_t [seq_modconf](#)
- uint8_t [seq_config](#)
- uint8_t [seq_sar_wait](#)
- uint8_t [seq_led_init](#)
- uint8_t [seq_freql](#)
- uint8_t [seq_freqh](#)
- uint8_t [seq1_freqdivl](#)
- uint8_t [seq1_freqdivh](#)
- uint8_t [seq2_freqdivl](#)
- uint8_t [seq2_freqdivh](#)
- uint8_t [mod1_seq1_sub_en](#)
- uint8_t [mod1_seq2_sub_en](#)
- uint8_t [mod2_seq1_sub_en](#)
- uint8_t [mod2_seq2_sub_en](#)
- uint8_t [seq1_mode_a](#)
- uint8_t [seq1_mode_b](#)
- uint8_t [seq2_mode](#)

5.27.1 Detailed Description

Register content for sequencer configuration

5.27.2 Field Documentation

5.27.2.1 irq_enable uint8_t as7056_57_config_seq_t::seq_regs::irq_enable

Content of register [AS7056_57_REGADDR_IRQ_ENABLE](#)

5.27.2.2 seq_sample uint8_t as7056_57_config_seq_t::seq_regs::seq_sample

Content of register [AS7056_57_REGADDR_SEQ_SAMPLE](#)

5.27.2.3 seq_sub_wait uint8_t as7056_57_config_seq_t::seq_regs::seq_sub_wait

Content of register [AS7056_57_REGADDR_SEQ_SUB_WAIT](#)

5.27.2.4 seq_modconf uint8_t as7056_57_config_seq_t::seq_regs::seq_modconf

Content of register [AS7056_57_REGADDR_SEQ_MODCONF](#)

5.27.2.5 seq_config uint8_t as7056_57_config_seq_t::seq_regs::seq_config

Content of register [AS7056_57_REGADDR_SEQ_CONFIG](#)

5.27.2.6 seq_sar_wait uint8_t as7056_57_config_seq_t::seq_regs::seq_sar_wait

Content of register [AS7056_57_REGADDR_SEQ_SAR_WAIT](#)

5.27.2.7 seq_led_init uint8_t as7056_57_config_seq_t::seq_regs::seq_led_init

Content of register [AS7056_57_REGADDR_SEQ_LED_INIT](#)

5.27.2.8 seq_freql uint8_t as7056_57_config_seq_t::seq_regs::seq_freql

Content of register [AS7056_57_REGADDR_SEQ_FREQL](#)

5.27.2.9 seq_freqh uint8_t as7056_57_config_seq_t::seq_regs::seq_freqh

Content of register [AS7056_57_REGADDR_SEQ_FREQH](#)

5.27.2.10 seq1_freqdivl uint8_t as7056_57_config_seq_t::seq_regs::seq1_freqdivl

Content of register [AS7056_57_REGADDR_SEQ1_FREQDIVL](#)

5.27.2.11 seq1_freqdivh uint8_t as7056_57_config_seq_t::seq_regs::seq1_freqdivh

Content of register [AS7056_57_REGADDR_SEQ1_FREQDIVH](#)

5.27.2.12 seq2_freqdivl uint8_t as7056_57_config_seq_t::seq_regs::seq2_freqdivl

Content of register [AS7056_57_REGADDR_SEQ2_FREQDIVL](#)

5.27.2.13 seq2_freqdivh uint8_t as7056_57_config_seq_t::seq_regs::seq2_freqdivh

Content of register [AS7056_57_REGADDR_SEQ2_FREQDIVH](#)

5.27.2.14 mod1_seq1_sub_en uint8_t as7056_57_config_seq_t::seq_regs::mod1_seq1_sub_en

Content of register [AS7056_57_REGADDR_MOD1_SEQ1_SUB_EN](#)

5.27.2.15 mod1_seq2_sub_en uint8_t as7056_57_config_seq_t::seq_regs::mod1_seq2_sub_en

Content of register [AS7056_57_REGADDR_MOD1_SEQ2_SUB_EN](#)

5.27.2.16 mod2_seq1_sub_en uint8_t as7056_57_config_seq_t::seq_regs::mod2_seq1_sub_en

Content of register [AS7056_57_REGADDR_MOD2_SEQ1_SUB_EN](#)

5.27.2.17 mod2_seq2_sub_en uint8_t as7056_57_config_seq_t::seq_regs::mod2_seq2_sub_en

Content of register [AS7056_57_REGADDR_MOD2_SEQ2_SUB_EN](#)

5.27.2.18 seq1_mode_a uint8_t as7056_57_config_seq_t::seq_regs::seq1_mode_a

Content of register [AS7056_57_REGADDR_SEQ1_MODE_A](#)

5.27.2.19 seq1_mode_b uint8_t as7056_57_config_seq_t::seq_regs::seq1_mode_b

Content of register [AS7056_57_REGADDR_SEQ1_MODE_B](#)

5.27.2.20 seq2_mode uint8_t as7056_57_config_seq_t::seq_regs::seq2_mode

Content of register [AS7056_57_REGADDR_SEQ2_MODE](#)

5.28 as7056_57_config_sinc_t::sinc_regs Struct Reference

Data Fields

- uint8_t [seq1_sinc_cfga](#)
- uint8_t [seq1_sinc_cfgb](#)
- uint8_t [seq1_sinc_cfgc](#)
- uint8_t [seq2_sinc_cfga](#)
- uint8_t [seq2_sinc_cfgb](#)
- uint8_t [seq2_sinc_cfgc](#)

5.28.1 Detailed Description

Register content for SINC filter configuration

5.28.2 Field Documentation

5.28.2.1 seq1_sinc_cfg_a `uint8_t as7056_57_config_sinc_t::sinc_regs::seq1_sinc_cfg_a`

Content of register [AS7056_57_REGADDR_SEQ1_SINC_CFGA](#)

5.28.2.2 seq1_sinc_cfg_b `uint8_t as7056_57_config_sinc_t::sinc_regs::seq1_sinc_cfg_b`

Content of register [AS7056_57_REGADDR_SEQ1_SINC_CFGB](#)

5.28.2.3 seq1_sinc_cfg_c `uint8_t as7056_57_config_sinc_t::sinc_regs::seq1_sinc_cfg_c`

Content of register [AS7056_57_REGADDR_SEQ1_SINC_CFGC](#)

5.28.2.4 seq2_sinc_cfg_a `uint8_t as7056_57_config_sinc_t::sinc_regs::seq2_sinc_cfg_a`

Content of register [AS7056_57_REGADDR_SEQ2_SINC_CFGA](#)

5.28.2.5 seq2_sinc_cfg_b `uint8_t as7056_57_config_sinc_t::sinc_regs::seq2_sinc_cfg_b`

Content of register [AS7056_57_REGADDR_SEQ2_SINC_CFGB](#)

5.28.2.6 seq2_sinc_cfg_c `uint8_t as7056_57_config_sinc_t::sinc_regs::seq2_sinc_cfg_c`

Content of register [AS7056_57_REGADDR_SEQ2_SINC_CFGC](#)

5.29 as7056_57_config_standby_t::standby_regs Struct Reference

Data Fields

- `uint8_t` [standby_on](#)
- `uint8_t` [standby_en1](#)
- `uint8_t` [standby_en2](#)
- `uint8_t` [standby_en3](#)
- `uint8_t` [standby_en4](#)
- `uint8_t` [standby_en5](#)
- `uint8_t` [standby_en6](#)
- `uint8_t` [standby_en7](#)

5.29.1 Detailed Description

Register content for STANDBY configuration

5.29.2 Field Documentation

5.29.2.1 standby_on uint8_t as7056_57_config_standby_t::standby_regs::standby_on

Content of register [AS7056_57_REGADDR_STANDBY_ON](#)

5.29.2.2 standby_en1 uint8_t as7056_57_config_standby_t::standby_regs::standby_en1

Content of register [AS7056_57_REGADDR_STANDBY_EN1](#)

5.29.2.3 standby_en2 uint8_t as7056_57_config_standby_t::standby_regs::standby_en2

Content of register [AS7056_57_REGADDR_STANDBY_EN2](#)

5.29.2.4 standby_en3 uint8_t as7056_57_config_standby_t::standby_regs::standby_en3

Content of register [AS7056_57_REGADDR_STANDBY_EN3](#)

5.29.2.5 standby_en4 uint8_t as7056_57_config_standby_t::standby_regs::standby_en4

Content of register [AS7056_57_REGADDR_STANDBY_EN4](#)

5.29.2.6 standby_en5 uint8_t as7056_57_config_standby_t::standby_regs::standby_en5

Content of register [AS7056_57_REGADDR_STANDBY_EN5](#)

5.29.2.7 standby_en6 uint8_t as7056_57_config_standby_t::standby_regs::standby_en6

Content of register [AS7056_57_REGADDR_STANDBY_EN6](#)

5.29.2.8 standby_en7 uint8_t as7056_57_config_standby_t::standby_regs::standby_en7

Content of register [AS7056_57_REGADDR_STANDBY_EN7](#)